

CSET340

Advanced Computer Vision and Video Analytics

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Overall Course Coordinator-

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Note : Any query related to course then first connect with overall course coordinator.

Camera Calibration

Camera calibration is a fundamental step in computer vision and image processing that acts as the bridge between the 2D world of images and the 3D world of physical space. (**Calibrate the camera means estimation of the projection matrix.**)

The primary **motivation** is to understand and mathematically model how a camera transforms a 3D scene into a 2D grid of pixels.

Without calibration, an image is just a collection of colored dots. With it, each dot becomes a precise geometric ray.

Why camera calibration is required –

1. Correction of lens distortion: Most camera lenses, especially wide-angle ones, do not project perfectly linear images. Light rays bend as they pass through the glass, leading to optical aberrations: like radial and tangential distortion.

2. Estimating Intrinsic and external parameters: To perform any measurement or 3D reconstruction, we need to know the "internal" characteristics of the camera. Like focal length, principal point, skew coefficient.

3. Geometric 3D Reconstruction:

The ultimate goal in many applications is to recover 3D information from 2D images.

Calibration is motivated by the need to solve:

Visual SLAM: Allowing robots or drones to map their surroundings and navigate through them accurately.

Stereo vision: Using two cameras to calculate depth (triangulation). You cannot find the distance to an object if you don't know the exact geometric relationship between the two cameras.

Camera Calibration:

From images to full metric reconstruction, we need **two things** -

Position and orientation of the camera with respect to the world coordinate frame.
(It referred to as **external parameter** of the camera)

How the camera maps the perspective projection points in the world onto its image plane.(it is referred to as **internal parameter** of the camera such as focal length, principal point.)

Determining the internal and external parameter of the camera is known as **camera calibration**

Topics that we are going to cover

Linear Camera Models: It's a forward imaging model which takes you from a point in 3D to its projection in the pixel in the image (2D).

Camera Calibration methods – We take a single picture of object of known geometry and that is all we need to fully calibrate the system or In other word to determine the projection matrix in full.

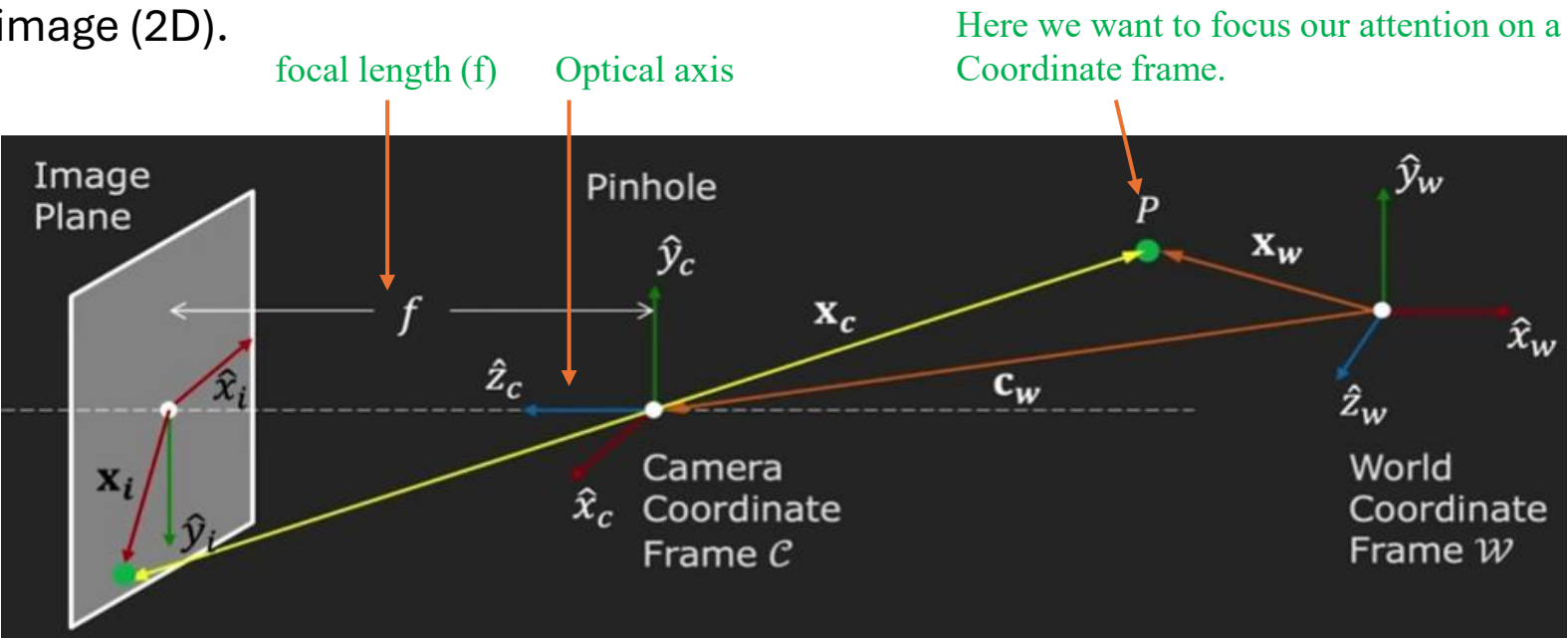
Extracting intrinsic and extrinsic matrices-

Once the projection matrix is calculated, we can figure out the external and internal parameter of the camera.

Example Application: Simple stereo

Camera Calibration:

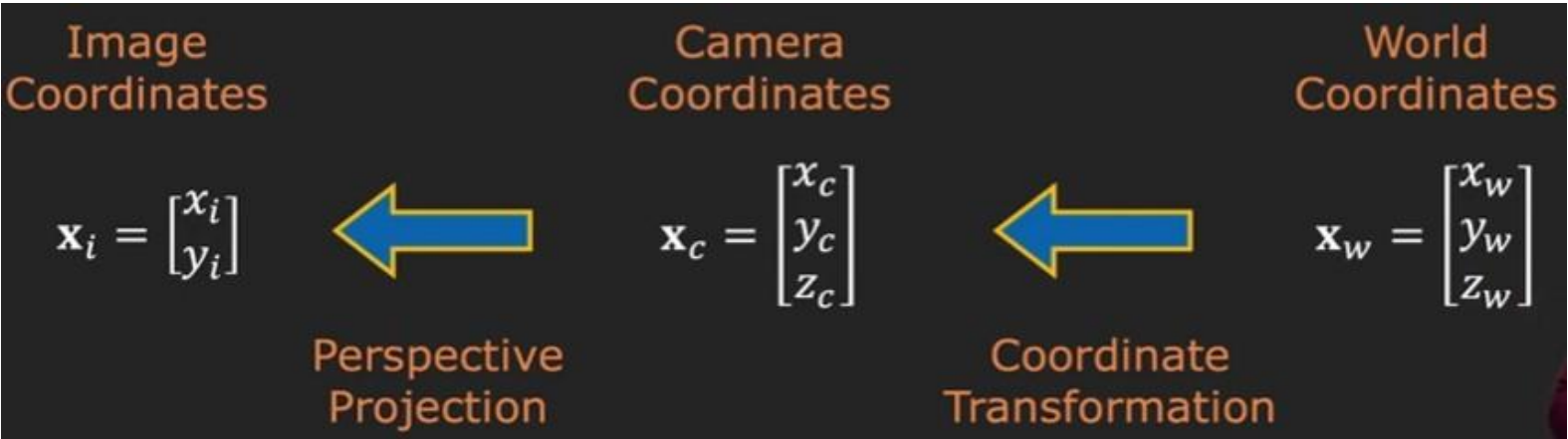
Linear Camera Models: It's a forward imaging model which takes you from a point in 3D to its projection in the pixel in the image (2D).



Here we want to focus our attention on a single point p in the world Coordinate frame.

Now, if we know the relative position and orientation of the Camera coordinate frame with respect to the world coordinate frame then we can write an expression that takes you all the way from the point p in the world coordinate frame to its projection X_i on the image plane.

This complete mapping is referred as forward imaging model, where it map from world coordinate to image coordinate.



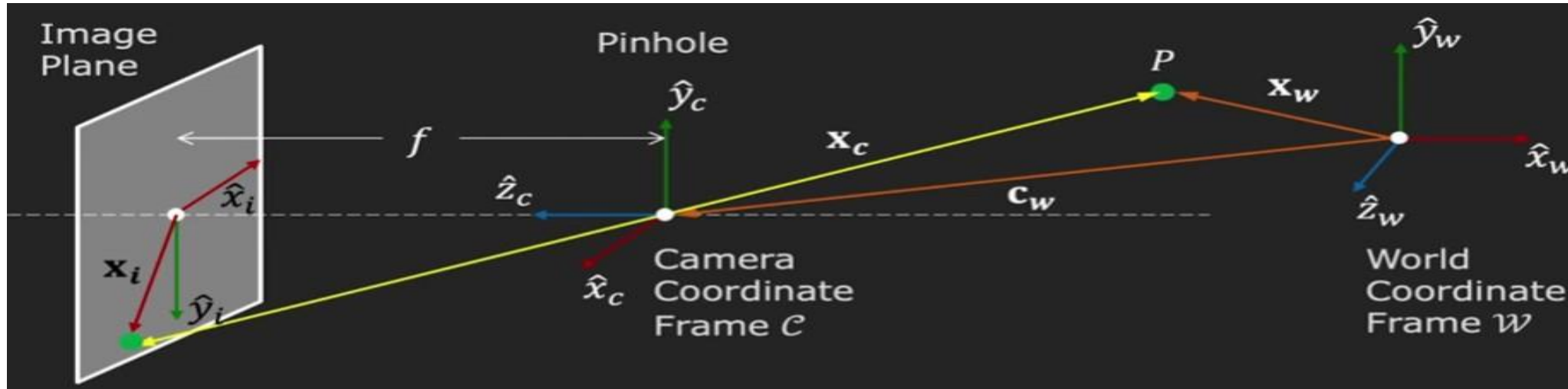
Here, we start with world coordinate frame and model its transformation to the camera coordinates frame (its 3D to 3D transformation).

Now, once we have camera coordinate then we can apply perspective projection to end up with image coordinates which is 3D to 2D coordinates on the image plane.

We are going to use it to develop comprehensive linear model for the camera.



Camera Calibration



Perspective Projection:

We know that: $\frac{x_i}{f} = \frac{x_c}{z_c}$ and $\frac{y_i}{f} = \frac{y_c}{z_c}$

Therefore: $x_i = f \frac{x_c}{z_c}$ and $y_i = f \frac{y_c}{z_c}$

Here, x_i and y_i are the coordinates of the projection of point P on to the image plane.

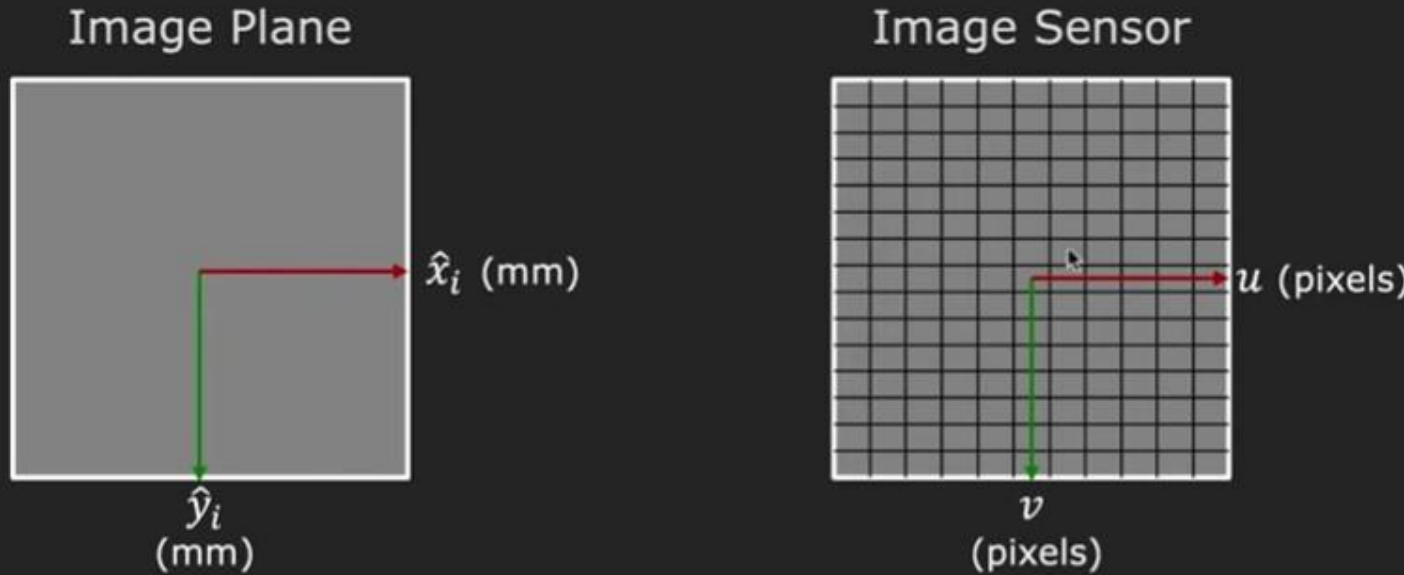
NOTE: Perspective projection is the process of projecting 3D scene or object onto a 2D plane or image.

Image Plane:

Lets assume we know the coordinate in millimeter. (Camera coordinate frame)

Image sensor:

In reality, what we have the image sensors which is used to capture the image and it has the pixels.



Pixels may be rectangular.

If m_x and m_y are the pixel densities (pixels/mm) in x and y directions, respectively, then pixel coordinates are:

$$u = m_x x_i = m_x f \frac{x_c}{z_c}$$

$$v = m_y y_i = m_y f \frac{y_c}{z_c}$$

Here, so we have the pixel coordinate (U,V) and we have to figure out the mapping from the image coordinate here in millimeters to pixel.

Here,

m_x = Number of pixels per millimeter in the x direction.

m_y = Number of pixels per millimeter in the y direction.

Now, we can say that the pixel coordinate of the projection of point P i.e. U and V-

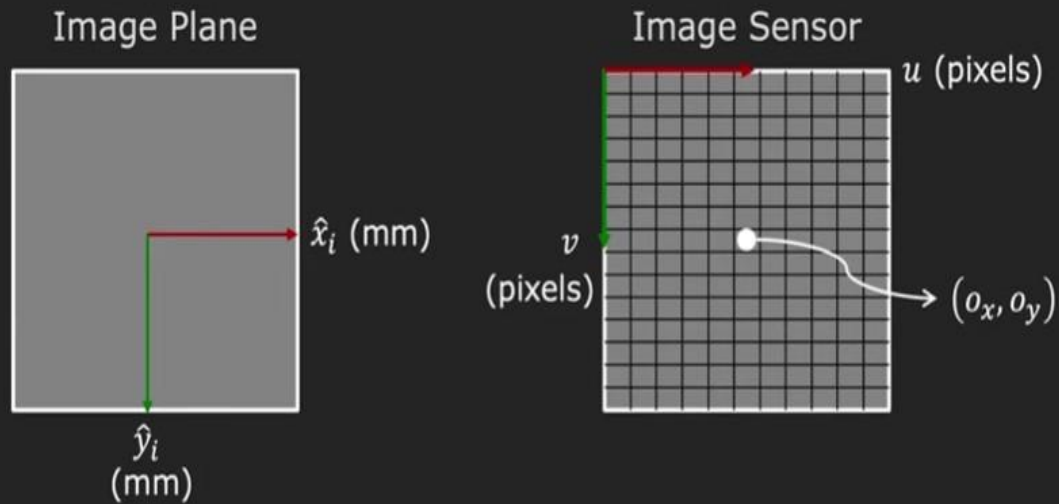
$U = m_x \times x_i$, $V = m_y \times y_i$

Here, you have gone from millimeter to pixel

Now, remember that this pixel density m_x and m_y are unknown they are the part of camera calibration process.

Camera Calibration

Here, There is one more thing to worry about which is thus far we have assumed that we know where the center of images (i.e. 0,0) principal point.



We usually treat the top-left corner of the image sensor as its origin (easier for indexing). If pixel (o_x, o_y) is the **Principle Point** where the optical axis pierces the sensor, then:

$$u = m_x f \frac{x_c}{z_c} + o_x \quad v = m_y f \frac{y_c}{z_c} + o_y$$

Here, O_x and O_y is also unknown for us. We can modify our perspective projection equation which include O_x and O_y .

$$u = m_x f \frac{x_c}{z_c} + o_x$$

$$v = m_y f \frac{y_c}{z_c} + o_y$$

$$u = f_x \frac{x_c}{z_c} + o_x$$

$$v = f_y \frac{y_c}{z_c} + o_y$$

where: $(f_x, f_y) = (m_x f, m_y f)$ are the focal lengths in pixels in the x and y directions.

(f_x, f_y, o_x, o_y) : **Intrinsic parameters** of the camera. They represent the **camera's internal geometry**.

But camera has only one focal length then why we used F_x and F_y .

In order to accommodate non equal pixel density in the x and y directions (pixel may be rectangular) we have different focal length in the x and y direction

Equations for perspective projection are **Non-Linear**.

It is convenient to express them as linear equations.

But how we go from nonlinear to a linear model where sometimes it is possible to do using **homogenous coordinates**.

Camera Calibration

Homogenous coordinates or representation

Homogenous coordinates are a system of coordinates used in the projective geometry; it allows to represent n-dimensional point to n + 1 point or numbers.

To convert from Cartesian to Homogeneous coordinates, we add an extra dimension (usually w).

•**2D Cartesian:** (x, y) to **Homogeneous:** (x, y, 1)

•**3D Cartesian:** (X, Y, Z) to **Homogeneous:** (X, Y, Z, 1)

•Reverse:

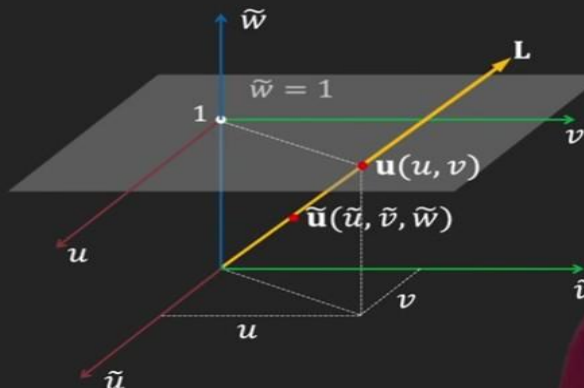
To go back, we divide the first n coordinates by the last one:

(x, y, w) to **Cartesian:** (x/w, y/w)

This representation is powerful because it treats points at infinity as valid coordinates (where w=0), which is essential for modeling how parallel lines appear to meet at a "vanishing point" in a camera image.

The **homogenous** representation of a 2D point $\mathbf{u} = (u, v)$ is a 3D point $\tilde{\mathbf{u}} = (\tilde{u}, \tilde{v}, \tilde{w})$. The third coordinate $\tilde{w} \neq 0$ is fictitious such that:

$$u = \frac{\tilde{u}}{\tilde{w}} \quad v = \frac{\tilde{v}}{\tilde{w}}$$



$$\mathbf{u} \equiv \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{w}u \\ \tilde{w}v \\ \tilde{w} \end{bmatrix} \equiv \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \tilde{\mathbf{u}}$$

The **homogenous** representation of a 3D point $\mathbf{x} = (x, y, z) \in \mathcal{R}^3$ is a 4D point $\tilde{\mathbf{x}} = (\tilde{x}, \tilde{y}, \tilde{z}, \tilde{w}) \in \mathcal{R}^4$. The fourth coordinate $\tilde{w} \neq 0$ is fictitious such that:

$$x = \frac{\tilde{x}}{\tilde{w}} \quad y = \frac{\tilde{y}}{\tilde{w}} \quad z = \frac{\tilde{z}}{\tilde{w}}$$

$$\mathbf{x} \equiv \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{w}x \\ \tilde{w}y \\ \tilde{w}z \\ \tilde{w} \end{bmatrix} \equiv \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \tilde{\mathbf{x}}$$

NOTE: Fictitious is generally used for scaling or normalization purpose.

Camera Calibration

Perspective projection equations:

$$u = f_x \frac{x_c}{z_c} + o_x \quad v = f_y \frac{y_c}{z_c} + o_y$$

Homogenous coordinates of (u, v) :

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \equiv \begin{bmatrix} z_c u \\ z_c v \\ z_c \end{bmatrix} = \begin{bmatrix} f_x x_c + z_c o_x \\ f_y y_c + z_c o_y \\ z_c \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

where: $(u, v) = (\tilde{u}/\tilde{w}, \tilde{v}/\tilde{w})$

Linear Model for Perspective Projection

Here, we can represent the pixel coordinates (u, v) as the homogenous coordinates.

We can further simplify it by a 3 x 4 matrix, which includes all the internal parameter of the camera multiply by the homogenous coordinates of 3-dimensional point defined in the camera coordinate frame X_c, Y_c and Z_c .

This tell us that we have linear model for perspective projection.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

Calibration Matrix: $K = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix}$

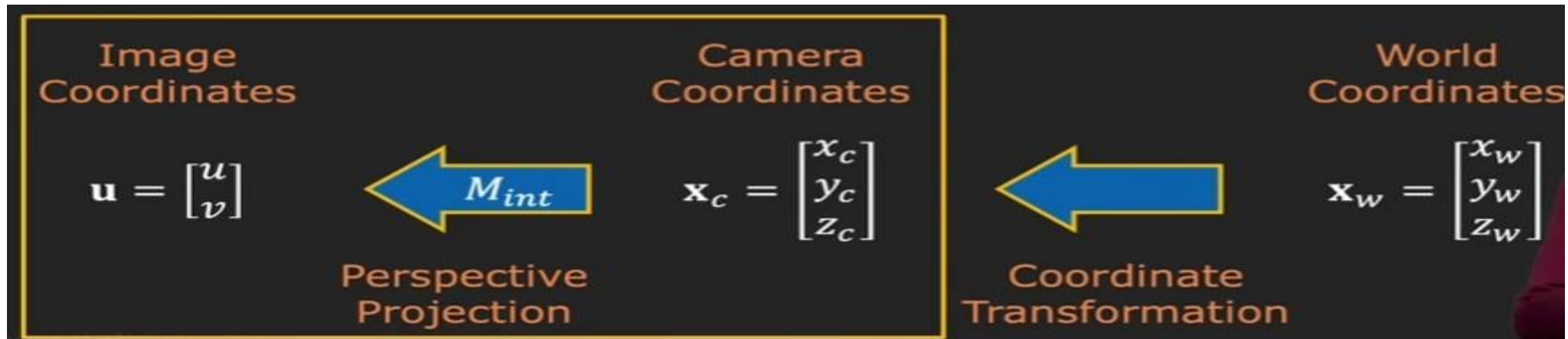
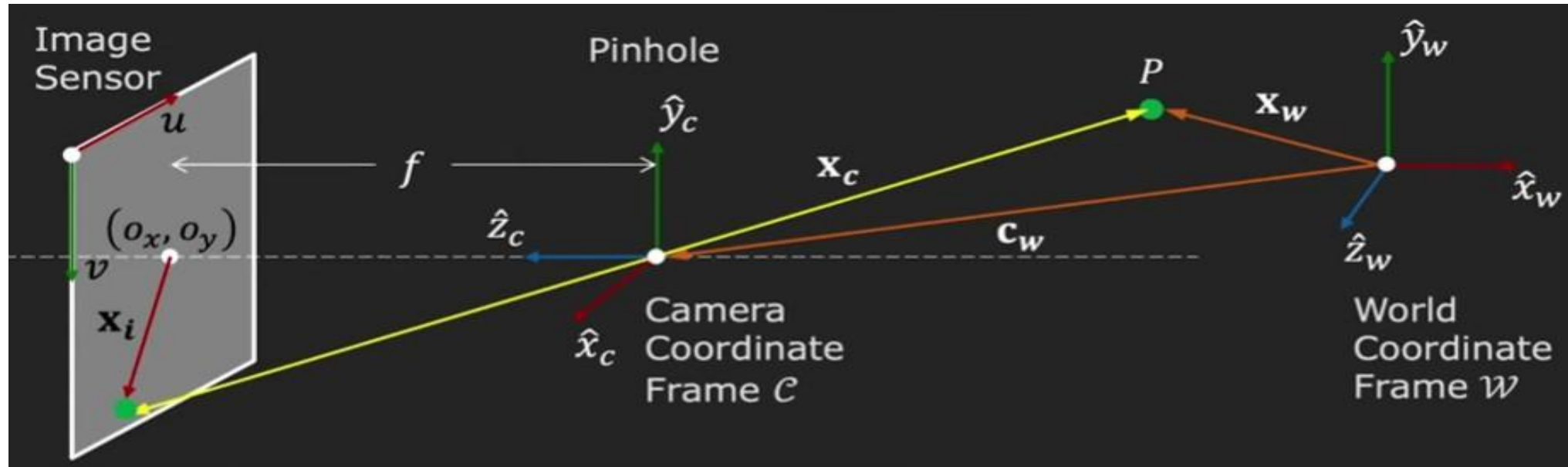
Intrinsic Matrix: $M_{int} = [K|0] = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Upper Right Triangular Matrix

$$\tilde{\mathbf{u}} = [K|0] \tilde{\mathbf{x}}_c = M_{int} \tilde{\mathbf{x}}_c$$

Mint - It takes you from a point homogenous coordinate representation of a point in the camera coordinate frame (3D) to its pixel coordinate in the image.

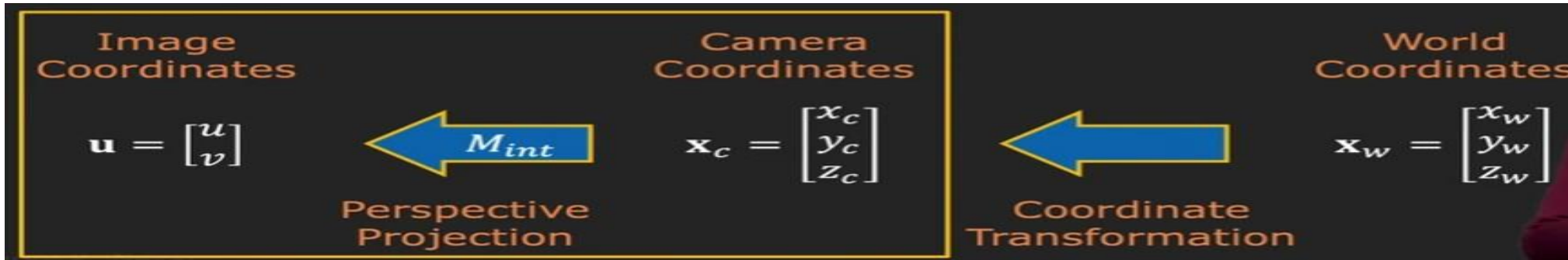
Camera Calibration



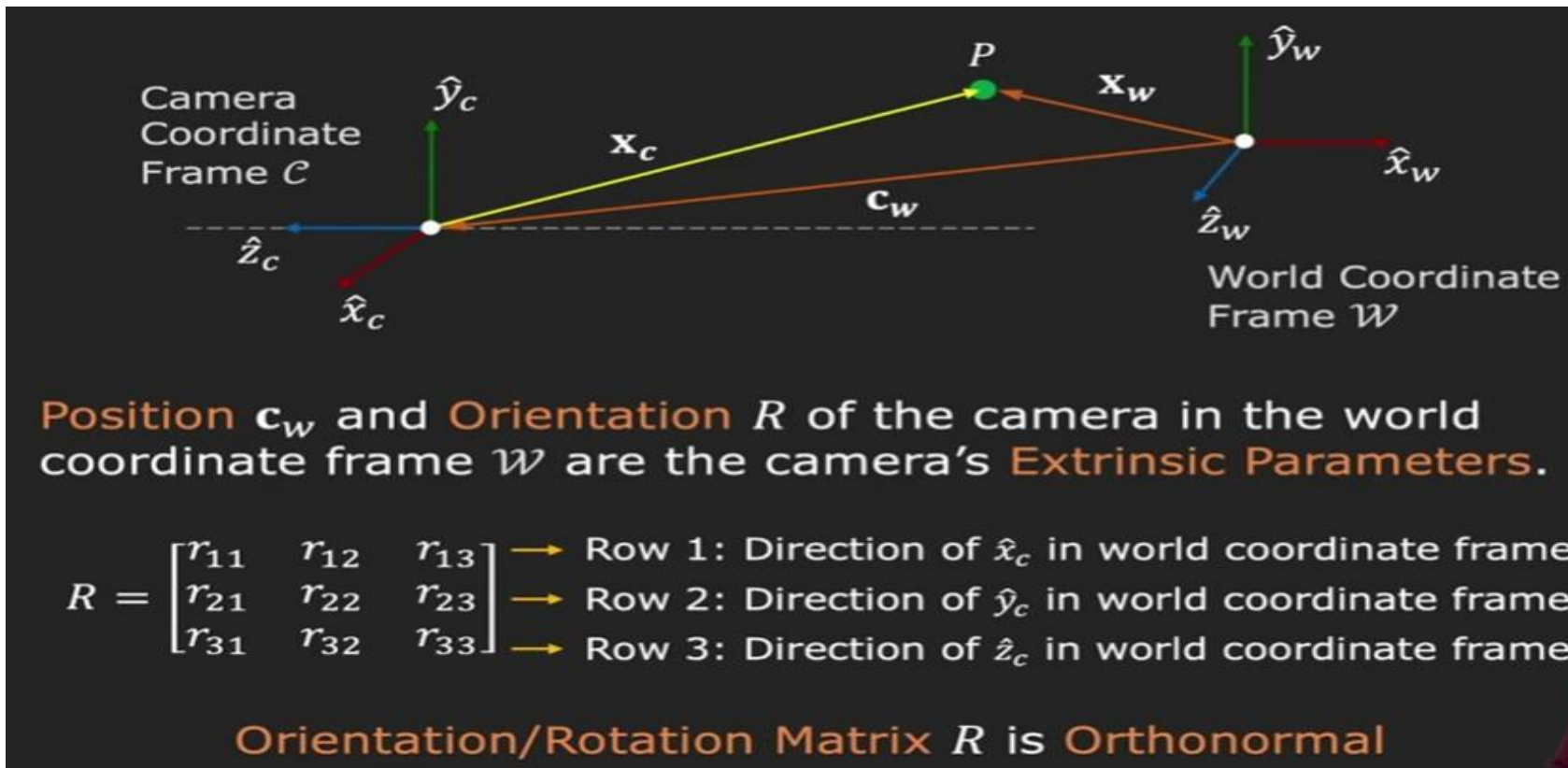
Here, we have done with perspective projection and ended with a linear model.

Next is, Mapping of a point from world coordinate to the camera coordinate.

Camera Calibration



Next is, Mapping of a point from world coordinate to the camera coordinate.



Extrinsic parameter:

Here, mapping is done from world coordinate frame (3D) to the camera coordinate frame (3D).

It can be done by using the position and orientation of the camera coordinate frame with respect to world coordinate frame.

The position is given by the vector ($\mathbf{c}_w$) and the orientation is given by the rotation matrix (R).

Interpretation:

First row of R :

Orientation of the vector $\mathbf{x}_c$ in the world coordinate frame is given by first row.

R is orthonormal matrix. What is orthonormal ?

Orthonormal Vectors: Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthonormal if and only if:

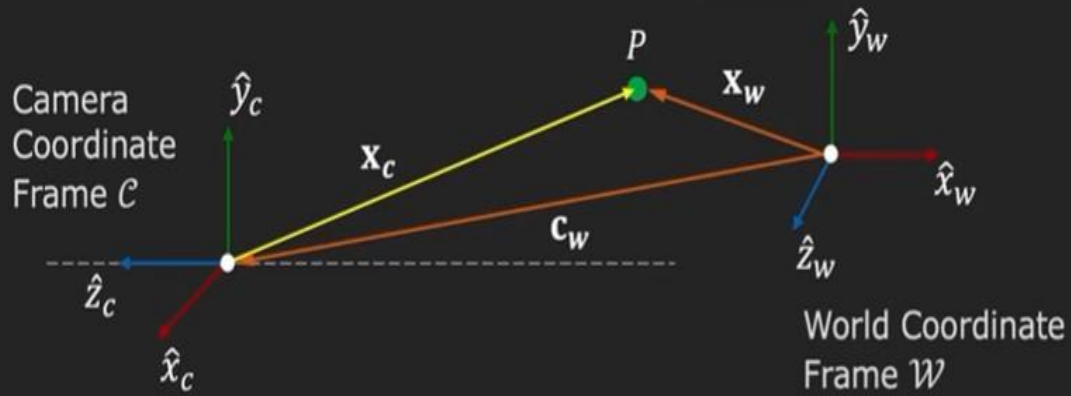
$$\begin{array}{ccc} \text{dot}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^T \mathbf{v} = 0 & \text{and} & \mathbf{u}^T \mathbf{u} = \mathbf{v}^T \mathbf{v} = 1 \\ \text{(Orthogonality)} & & \text{(Unit length)} \end{array}$$

Example: The x -, y - and z -axes of $\mathbb{R}^3$ Euclidean space

Orthonormal Matrix: A square matrix R whose row (or column) vectors are orthonormal. For such a matrix:

$$R^{-1} = R^T \quad R^T R = R R^T = I$$

Camera Calibration



Given the **extrinsic parameters** (R, c_w) of the camera, the camera-centric location of the point P in the world coordinate frame is:

$X_c = X_w - C_w$ in the world coordinate frame. When you multiply it with rotation matrix (R) then you have X_c in the camera coordinate frame.

$$\mathbf{x}_c = R(\mathbf{x}_w - \mathbf{c}_w) = R\mathbf{x}_w - R\mathbf{c}_w = R\mathbf{x}_w + \mathbf{t}$$

$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Here t is the translation vector.

Rewriting using homogenous coordinates:

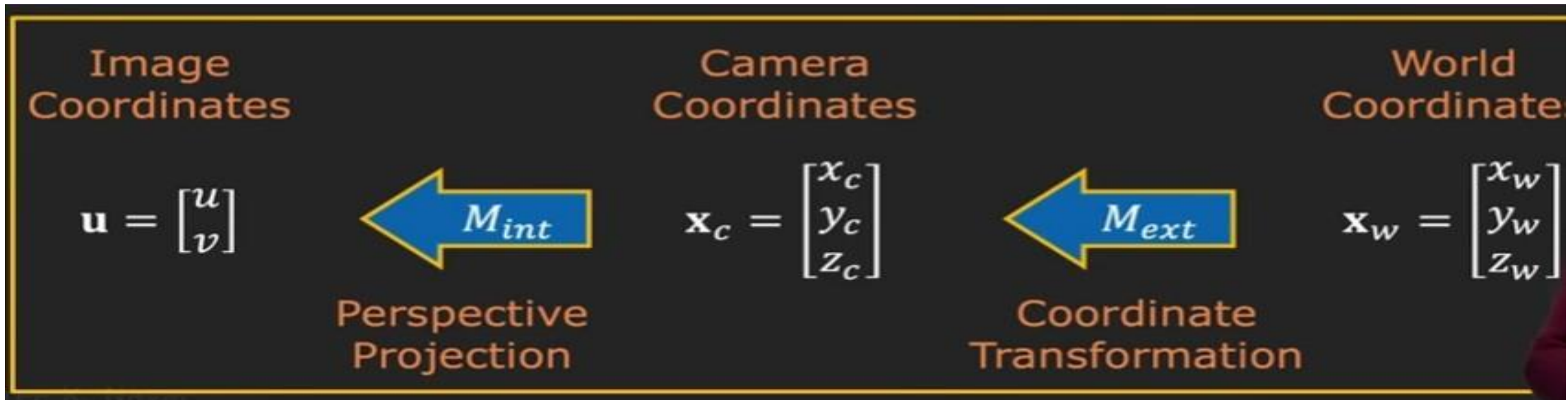
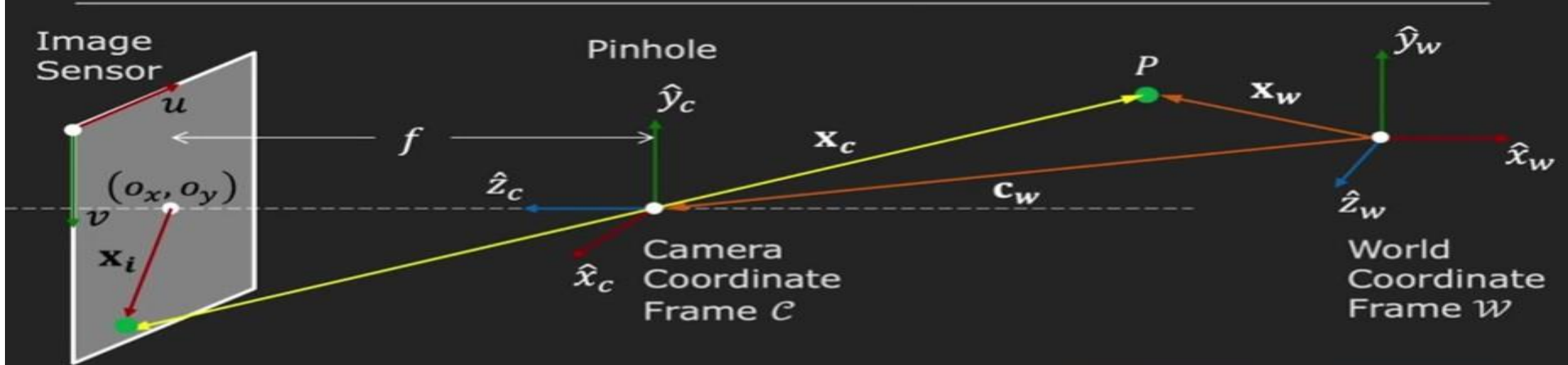
$$\tilde{\mathbf{x}}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

Extrinsic Matrix: $M_{ext} = \begin{bmatrix} R_{3 \times 3} & \mathbf{t} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$\tilde{\mathbf{x}}_c = M_{ext} \tilde{\mathbf{x}}_w$$

Camera Calibration

Forward Imaging Model: 3D to 2D



Camera Calibration

Projection Matrix P

Camera to Pixel

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}} = M_{int} \tilde{\mathbf{x}}_c$$

World to Camera

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{x}}_c = M_{ext} \tilde{\mathbf{x}}_w$$

Combining the above two equations, we get the full projection matrix P :

$$\tilde{\mathbf{u}} = M_{int} M_{ext} \tilde{\mathbf{x}}_w = P \tilde{\mathbf{x}}_w$$

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

Here, both the intrinsic and extrinsic matrix are combined and get a full projection matrix p .

Projection matrix maps a point p in the world coordinate frame to a pixel in the image.

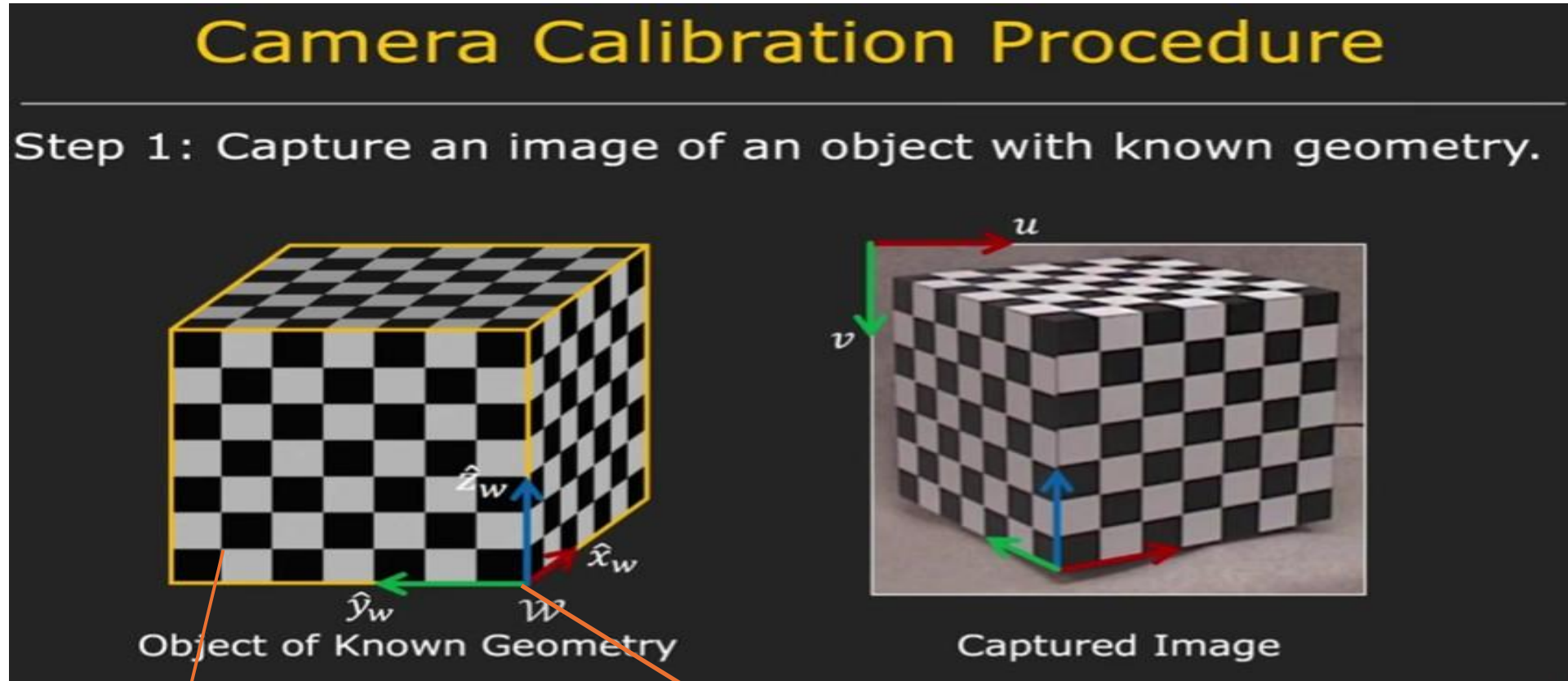
Projection matrix map from millimeter to pixel.

NOTE

If we want to perform the camera calibration, then we need to know about all these 12 parameter or the numbers of the projection matrix P .

Camera Calibration

Now, we have linear camera model which is given by the projection matrix. We are ready to calibrate the camera which essentially means estimate the projection matrix.



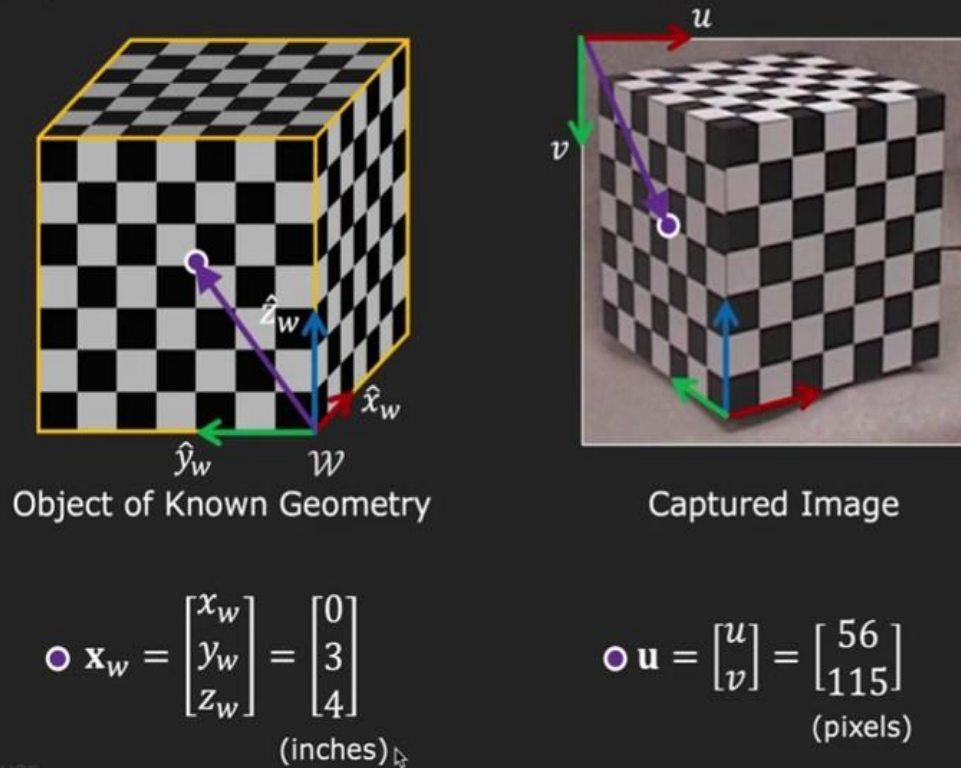
We know everything about this cube – like dimensions, location of every point on this cube, every visual feature in 3D.

Now world coordinate frame is placed on one of the corner of this cube, all the points are going to be measured with respect to this coordinate frame.

Camera Calibration

Step-2 Identify the correspondences between 3D scene points in the world coordinate frame and image points in the image plane.

Step 2: Identify correspondences between 3D scene points and image points.



Here, you can select all the other points. By selecting all the points, you will get the set of corresponding points between 3D to 2D.

If you get or select one of these points

Step 3: For each corresponding point i in scene and image:

$$\underbrace{\begin{bmatrix} u^{(i)} \\ v^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}} \equiv \underbrace{\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} x_w^{(i)} \\ y_w^{(i)} \\ z_w^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}}$$

Expanding the matrix as linear equations:

$$u^{(i)} = \frac{p_{11}x_w^{(i)} + p_{12}y_w^{(i)} + p_{13}z_w^{(i)} + p_{14}}{p_{31}x_w^{(i)} + p_{32}y_w^{(i)} + p_{33}z_w^{(i)} + p_{34}}$$

$$v^{(i)} = \frac{p_{21}x_w^{(i)} + p_{22}y_w^{(i)} + p_{23}z_w^{(i)} + p_{24}}{p_{31}x_w^{(i)} + p_{32}y_w^{(i)} + p_{33}z_w^{(i)} + p_{34}}$$

Camera Calibration

Step 4: Rearranging the terms

$$\begin{bmatrix}
 x_w^{(1)} & y_w^{(1)} & z_w^{(1)} & 1 & 0 & 0 & 0 & 0 & -u_1 x_w^{(1)} & -u_1 y_w^{(1)} & -u_1 z_w^{(1)} & -u_1 \\
 0 & 0 & 0 & 0 & x_w^{(1)} & y_w^{(1)} & z_w^{(1)} & 1 & -v_1 x_w^{(1)} & -v_1 y_w^{(1)} & -v_1 z_w^{(1)} & -v_1 \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 x_w^{(i)} & y_w^{(i)} & z_w^{(i)} & 1 & 0 & 0 & 0 & 0 & -u_i x_w^{(i)} & -u_i y_w^{(i)} & -u_i z_w^{(i)} & -u_i \\
 0 & 0 & 0 & 0 & x_w^{(i)} & y_w^{(i)} & z_w^{(i)} & 1 & -v_i x_w^{(i)} & -v_i y_w^{(i)} & -v_i z_w^{(i)} & -v_i \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 x_w^{(n)} & y_w^{(n)} & z_w^{(n)} & 1 & 0 & 0 & 0 & 0 & -u_n x_w^{(n)} & -u_n y_w^{(n)} & -u_n z_w^{(n)} & -u_n \\
 0 & 0 & 0 & 0 & x_w^{(n)} & y_w^{(n)} & z_w^{(n)} & 1 & -v_n x_w^{(n)} & -v_n y_w^{(n)} & -v_n z_w^{(n)} & -v_n
 \end{bmatrix}
 \begin{bmatrix}
 p_{11} \\
 p_{12} \\
 p_{13} \\
 p_{14} \\
 p_{21} \\
 p_{22} \\
 p_{23} \\
 p_{24} \\
 p_{31} \\
 p_{32} \\
 p_{33} \\
 p_{34}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0
 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_A$
 $\underbrace{\hspace{10em}}_{\mathbf{p}}$

Known
 Unknown

Step 5: Solve for $\mathbf{p}$

$$A \mathbf{p} = \mathbf{0}$$

What is the least square solution for $\mathbf{P}$?

Least Squares Solution for P

Option 1: Set scale so that: $p_{34} = 1$

Option 2: Set scale so that: $\|\mathbf{p}\|^2 = 1$

We want $A\mathbf{p}$ as close to 0 as possible and $\|\mathbf{p}\|^2 = 1$:

$$\min_{\mathbf{p}} \|A\mathbf{p}\|^2 \text{ such that } \|\mathbf{p}\|^2 = 1$$

$$\min_{\mathbf{p}} (\mathbf{p}^T A^T A \mathbf{p}) \text{ such that } \mathbf{p}^T \mathbf{p} = 1$$

Define Loss function $L(\mathbf{p}, \lambda)$:

$$L(\mathbf{p}, \lambda) = \mathbf{p}^T A^T A \mathbf{p} - \lambda(\mathbf{p}^T \mathbf{p} - 1)$$

Constrained Least Squares Solution

Taking derivatives of $L(\mathbf{p}, \lambda)$ w.r.t $\mathbf{p}$: $2A^T A \mathbf{p} - 2\lambda \mathbf{p} = 0$

$$A^T A \mathbf{p} = \lambda \mathbf{p}$$

Eigenvalue Problem

Eigenvector $\mathbf{p}$ with **smallest eigenvalue** λ of matrix $A^T A$ minimizes the loss function $L(\mathbf{p})$.

Rearrange solution $\mathbf{p}$ to form the projection matrix P .

Camera Calibration

Extracting intrinsic and extrinsic parameters -

We know the projection matrix P and we can decompose it into internal and external parameters.

Here, our projection matrix P is multiplication of the intrinsic matrix M_{int} and external matrix M_{ext} .

We know that:

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

M_{int} M_{ext}

That is:

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = KR$$

Given that K is an **Upper Right Triangular** matrix and R is an **Orthonormal** matrix, it is possible to uniquely “decouple” K and R from their product using “**QR factorization**”.

If you have matrix which is product of upper triangular matrix and orthonormal matrix. You can use very powerful tool like QR factorization to compute the two matrices from their product.

We know that:

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

That is:

$$\begin{bmatrix} p_{14} \\ p_{24} \\ p_{34} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} = K\mathbf{t}$$

Therefore:

$$\mathbf{t} = K^{-1} \begin{bmatrix} p_{14} \\ p_{24} \\ p_{34} \end{bmatrix}$$

Another parameter translation is also calculated with the help of the last column of projection matrix and calibration matrix K .

CSET340

Advanced Computer Vision and Video Analytics

13th April. to 17th April. 2026

Overall Course Coordinator-

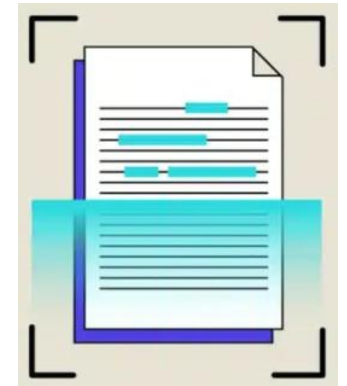
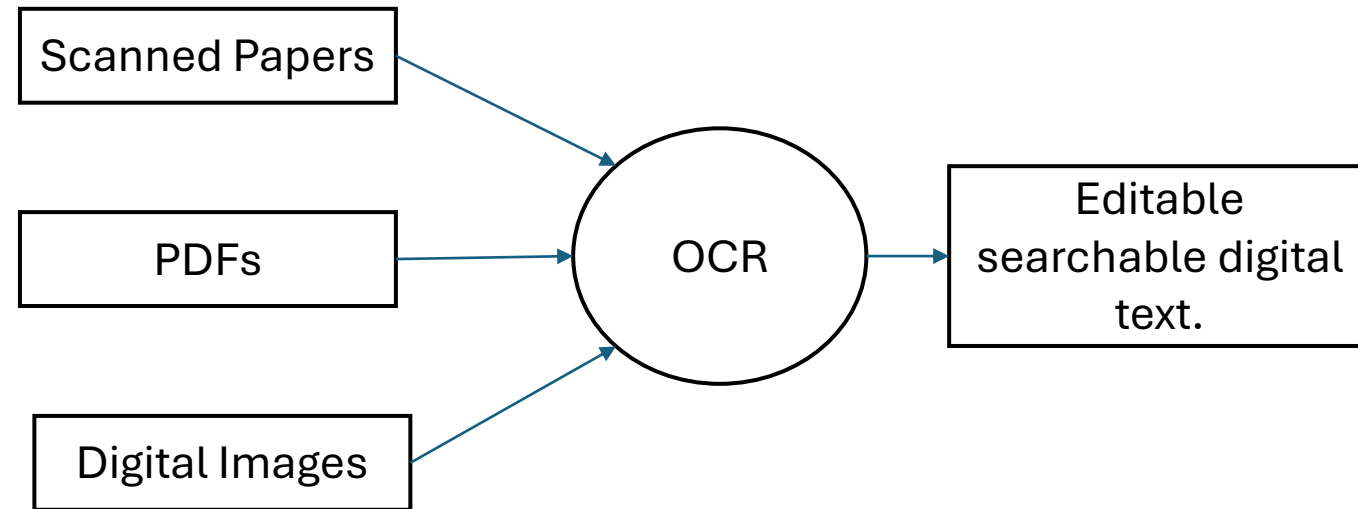
Dr. Gaurav Kumar Dashondhi

Gaurav.dashondhi@bennett.edu.in

Note : Any query related to course then first connect with overall course coordinator.

Optical Character Recognition (OCR): Definition & Motivation

OCR: It converts different type of documents including scanned papers, PDFs and digital images into editable searchable digital text.



Document Scanning

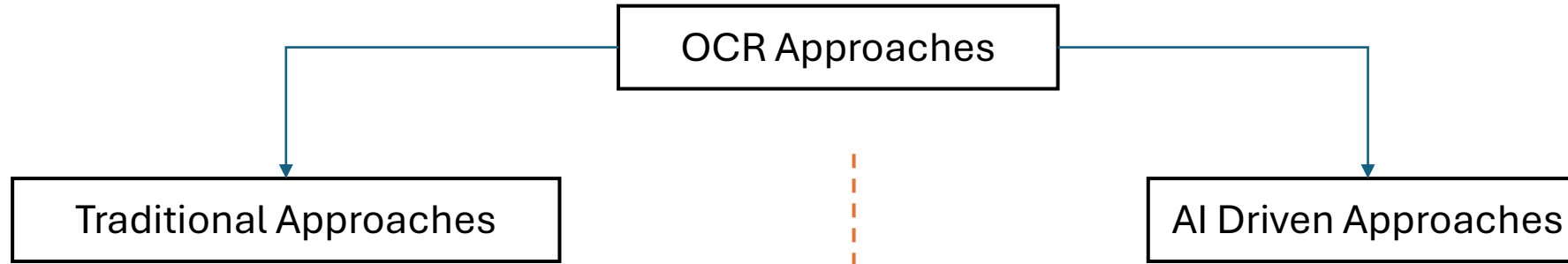
Key application of OCR:

Document digitization: Converting historical documents into searchable digital archives.

Automated data entry: Automatically extracting the key information from structured documents like invoices, receipts etc.

Accessibility: Converting printed text documents to digital format that can be used with text to speech software for the people who have reading difficulties.

Optical Character Recognition: Approaches



Traditional OCR relies on:

Template matching for character recognition.

Pre-defined rules and patterns.

Basic image processing techniques.

Limited accuracy with complex layouts.

Modern AI-driven OCR relies on:

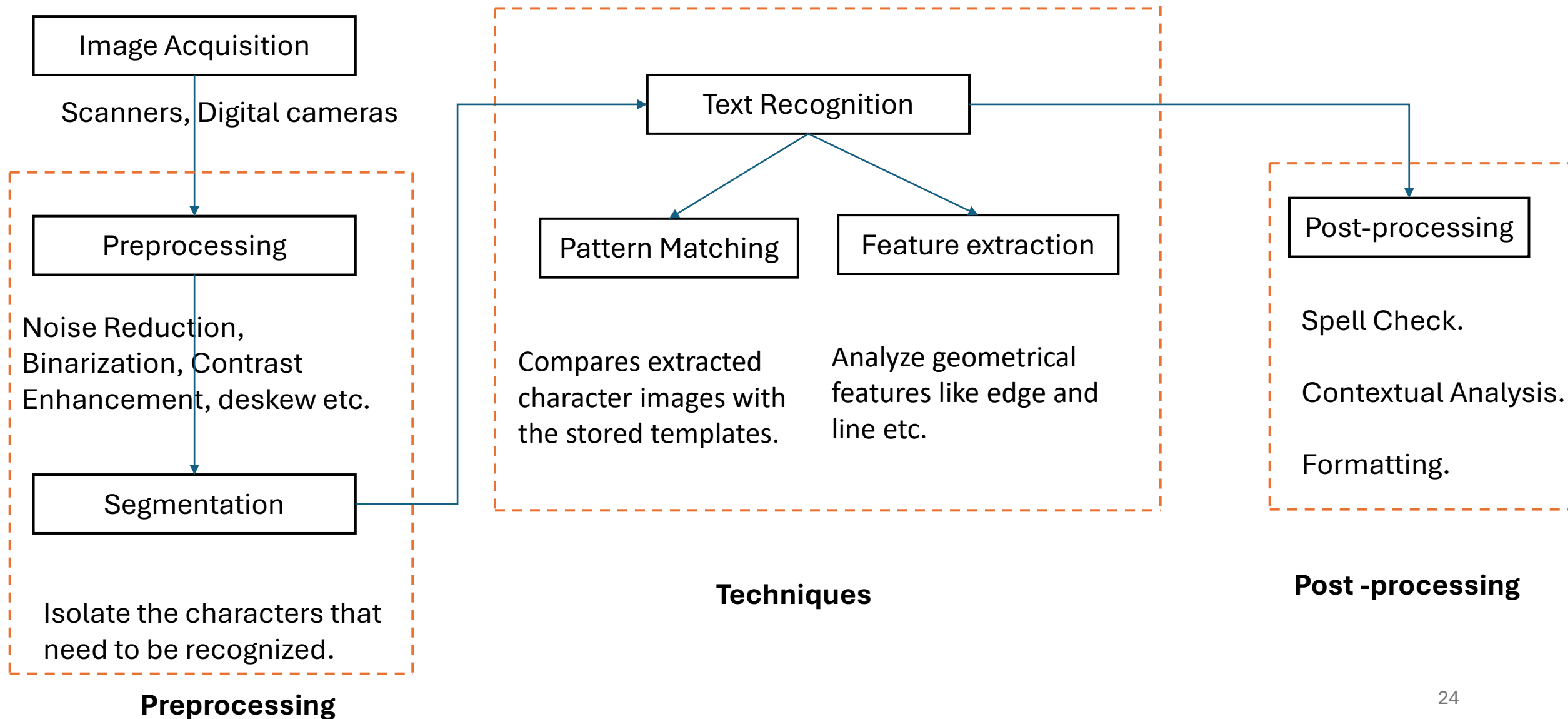
Deep learning algorithms: Continuous improvement through training.

Computer vision: Advanced image understanding.

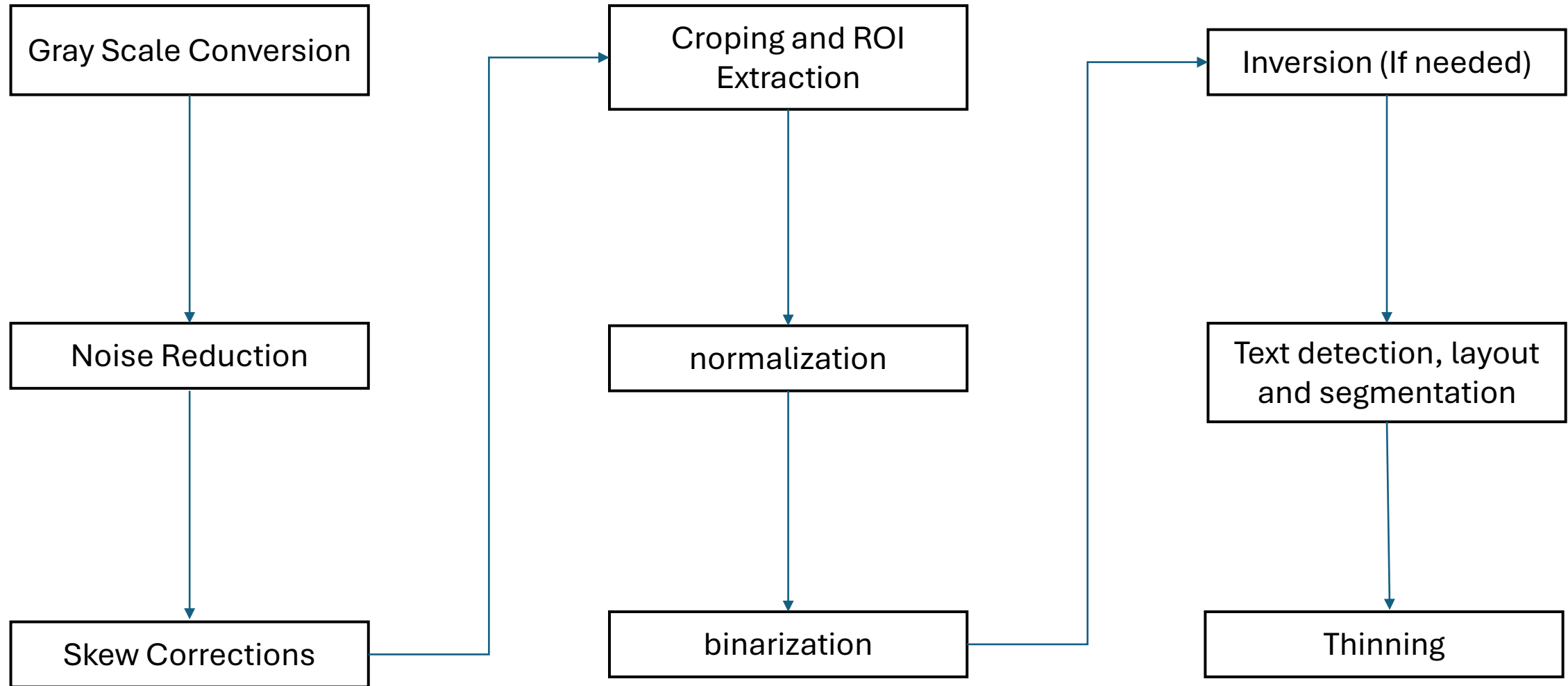
Natural Language Processing: Context-aware text interpretation.

Adaptive learning: Improves accuracy over time.

OCR: Text Recognition : Flow Chart



OCR: Preprocessing



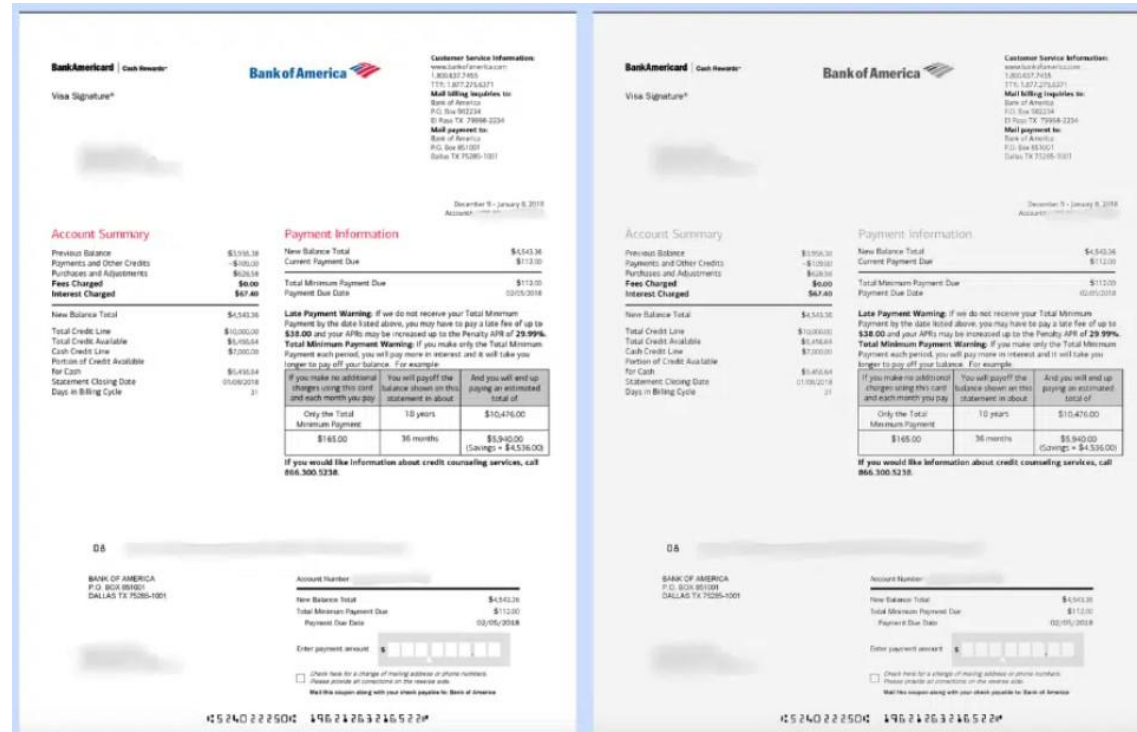
Note: Based on the requirements, preprocessing step sequence may change.

OCR: Preprocessing

1. Grayscale Conversion

- Goal:** Change color images to grayscale images.
- Techniques:** Simple averaging or Luminance method.
- Why:** Removes unnecessary color information, reduces complexity, and enhances further binarization or thresholding.

Color Image



Binarization or BW image

OCR: Preprocessing

2. Noise Reduction

- **Goal:** Uneven illumination, Remove specks, smudges, or background noise that can look like real text to an OCR engine.
- **Techniques:** Median filtering, morphological operations (like erosion/dilation), and smoothing.
Uneven illumination can be removed by contrast limited adaptive histogram equalization.
- **Why:** Cleaner images help the OCR systems to avoid false positives and reduce errors in character recognition.

Image with Noise

Bank of America | Cash Rewards

Customer Service Information:
www.bankofamerica.com
1-800-872-7428
TTY: 1-877-275-5371

Mail billing inquiries to:
Bank of America
P.O. Box 982336
Dallas, TX 75298-2336

Mail payment to:
Bank of America
P.O. Box 951001
Dallas, TX 75295-1001

December 9 - January 8, 2018
Account: ****

Account Summary

Previous Balance	\$1,555.36
Payments and Other Credits	-\$128.00
Purchases and Adjustments	\$0.00
Fees Charged	\$0.00
Interest Charged	\$67.40

New Balance Total \$1,484.36

Total Credit Line \$12,000.00
Total Credit Available \$7,500.00
Cash Credit Line \$4,450.00
Portion of Credit Available for Cash \$1,082.018
Statement Closing Date 01/08/2018
Days in Billing Cycle 31

Payment Information

New Balance Total	\$1,484.36
Current Payment Due	\$1,120.00
Total Minimum Payment Due	\$1,120.00
Payment Due Date	02/05/2018

Late Payment Warning: If we do not receive your Total Minimum Payment by the date listed above, you may have to pay a late fee of up to \$18.00 and your APRs may be increased up to the Penalty APR of 29.99%.

Total Minimum Payment Warning: If you make only the Total Minimum Payment each period, you will pay more in interest and it will take you longer to pay off your balance. For example:

If you make no additional charges using this card and each month you pay only the Total Minimum Payment	You will pay off the balance shown on this statement in about 10 years	And you will end up paying an estimated total of \$10,476.00
\$165.00	36 months	\$5,940.00 (Savings = \$4,536.00)

If you would like information about credit counseling services, call 866.300.5238.

08

BANK OF AMERICA
P.O. BOX 951001
DALLAS, TX 75295-1001

Account Number

New Balance Total \$1,484.36
Total Minimum Payment Due \$1,120.00
Payment Due Date 02/05/2018

Enter payment amount \$

☐ Check box for a change of billing address or phone numbers. Please provide all corrections on the reverse side.

Mail this coupon along with your check payable to Bank of America.

45240222501 149621263246522*

Image without Noise

Bank of America | Cash Rewards

Customer Service Information:
www.bankofamerica.com
1-800-872-7428
TTY: 1-877-275-5371

Mail billing inquiries to:
Bank of America
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Total Minimum Payment Warning: If you make only the Total Minimum Payment each period, you will pay more in interest and it will take you longer to pay off your balance. For example:

If you make no additional charges using this card and each month you pay only the Total Minimum Payment	You will pay off the balance shown on this statement in about 10 years	And you will end up paying an estimated total of \$10,476.00
\$165.00	36 months	\$5,940.00 (Savings = \$4,536.00)

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Enter payment amount \$

☐ Check box for a change of billing address or phone numbers. Please provide all corrections on the reverse side.

Mail this coupon along with your check payable to Bank of America.

45240222501 149621263246522*

OCR: Preprocessing

Steps in contrast limited adaptive histogram equalization (CLAHE).

Image Tiling or Partitioning

The input image is divided into a grid of small, non-overlapping, and equally sized **tiles**. The contrast enhancement is performed independently for each of these tiles.

Histogram computation and clipping or contrast limiting

For each tile, the **pixel intensity histogram** is computed.

Contrast Limiting: A clip limit is applied to each histogram bin. This limit, usually determined by the distribution of the histogram and a user-defined parameter, it prevents any single gray-level bin from having too high a count.

Any pixel count in a histogram bin that **exceeds the clip limit** is **clipped** (or truncated) to that limit.

The total number of clipped pixels (the **excess**) is then **redistributed evenly** among all the histogram bins. This ensures that the overall brightness of the tile is maintained. This step is what distinguishes CLAHE from standard AHE and prevents noise over-amplification.

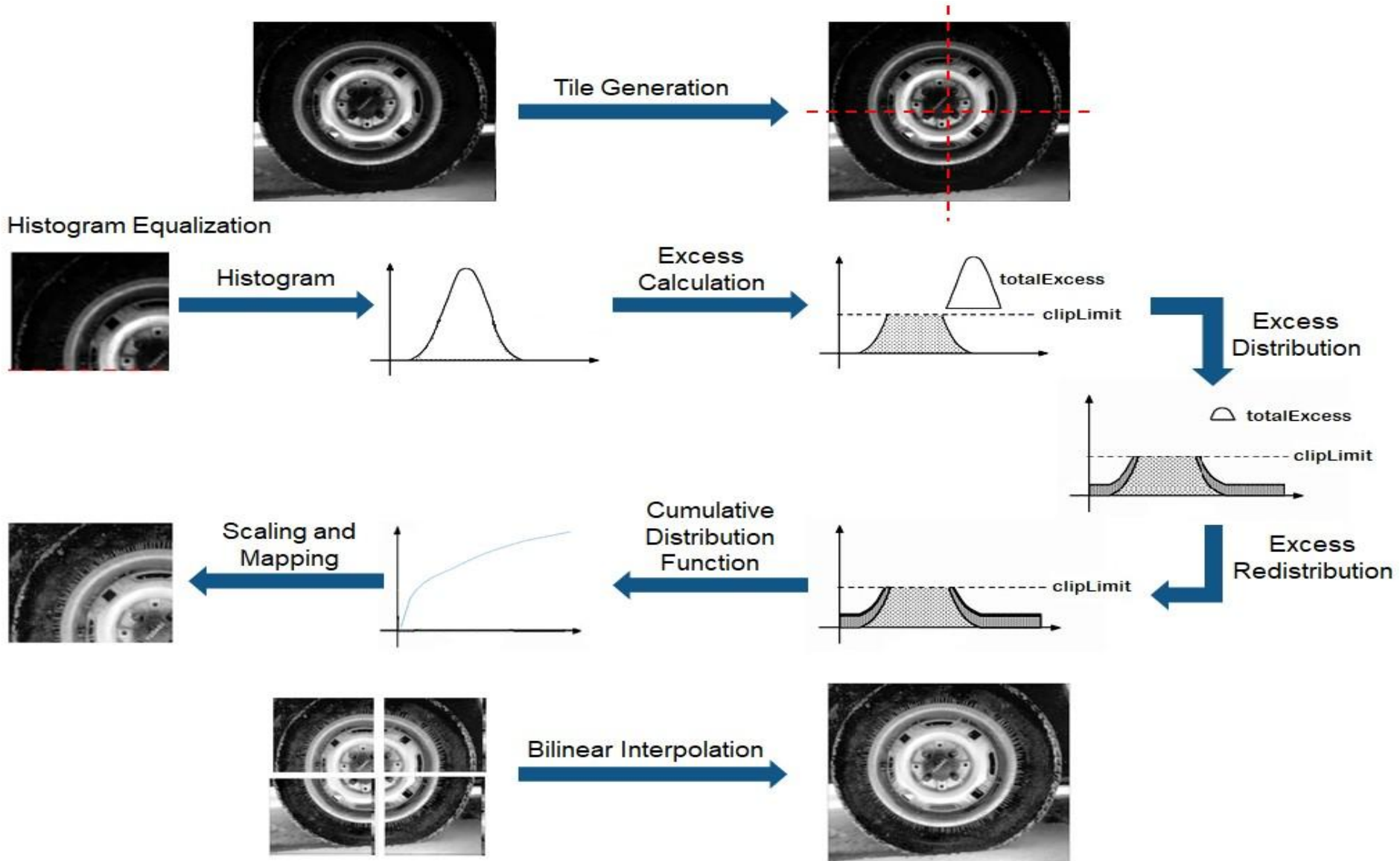
Local histogram equalization

A local histogram equalization has been performed, after the histogram has been clipped and redistributed.

Image Reconstruction via Bilinear Interpolation

OCR: Preprocessing

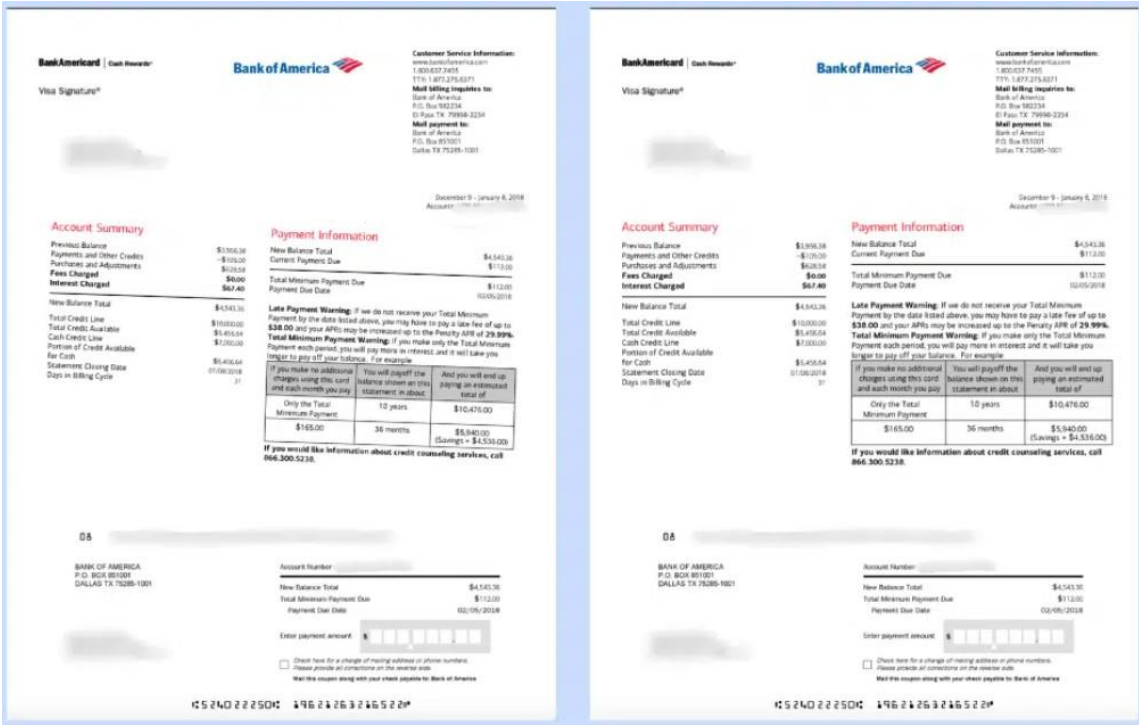
Steps in contrast limited adaptive histogram equalization (CLAHE).



Optical Character Recognition: Flow diagram detail

3. Skew Correction

- Goal:** Detects and corrects any tilt or rotation in the scanned image.
- Techniques:** Hough Transform etc.
- Why:** Even slight tilts cause major recognition problems, so aligning lines horizontally is crucial for maximal accuracy.



OCR: Preprocessing

Skew Correction Techniques



Hough Transform: Detects lines in edge maps; dominant line orientation reflects about skew.

Skew angle calculation is the process of estimating the rotation angle of the text baseline in a document image that causes the document's skew.

- The image is rotated by the detected angle in the opposite direction.
- Interpolation methods (e.g., bilinear) are used to prevent image artifacts.

4. Cropping and Region of Interest (ROI) Extraction

- **Goal:** Focus on relevant parts of the image, removing borders, graphics, or unrelated regions.
- **Techniques:** Manual/automatic cropping, contour/box detection around text.
- **Why:** Reduces distractions and keeps computation efficient and focused on the text content.

OCR: Preprocessing

Original Image



The image shows a full-page credit card statement from Bank of America. It includes the bank's logo, account information, a table of account summary, a table of payment information, and a table of late payment warnings. The text is in a standard sans-serif font, and the layout is clean and professional.

Account Summary	
Previous Balance	\$1,795.00
Payments and Other Credits	\$0.00
Purchases and Adjustments	\$0.00
Fees Charged	\$0.00
Interest Charged	\$61.45
New Balance Total	\$1,856.45
Total Credit Line	\$10,000.00
Total Credit Available	\$8,143.55
Cash Credit Line	\$7,000.00
Portion of Credit Available for Cash	\$1,400.00
Statement Closing Date	01/09/2018
Days in Billing Cycle	31

Payment Information	
New Balance Total	\$1,856.45
Current Payment Due	\$112.00
Total Minimum Payment Due	\$112.00
Payment Due Date	02/05/2018

Late Payment Warning	
If you do not receive your Total Minimum Payment by the date listed above, you may have to pay a late fee of up to \$35.00 and your APR may be increased up to the Penalty APR of 29.99%.	
Total Minimum Payment Warning: If you make only the Total Minimum Payment each period, you will pay more in interest and it will take you longer to pay off your balance. For example:	
If you make no additional charges using this card and only make the minimum payment:	If you pay off the balance shown on this statement in about 18 years
Only the Total Minimum Payment	\$112.00
	\$1,942.00 (Starting at \$1,856.45)

Cropped Image or ROI



The image is a cropped version of the original statement, focusing on the 'Account Summary' and 'Payment Information' sections. It removes the header, footer, and the 'Late Payment Warning' section.

Account Summary	
Previous Balance	\$1,795.00
Payments and Other Credits	\$0.00
Purchases and Adjustments	\$0.00
Fees Charged	\$0.00
Interest Charged	\$61.45
New Balance Total	\$1,856.45
Total Credit Line	\$10,000.00
Total Credit Available	\$8,143.55
Cash Credit Line	\$7,000.00
Portion of Credit Available for Cash	\$1,400.00
Statement Closing Date	01/09/2018
Days in Billing Cycle	31

Payment Information	
New Balance Total	\$1,856.45
Current Payment Due	\$112.00
Total Minimum Payment Due	\$112.00
Payment Due Date	02/05/2018

5.Normalization

- Goal:** Standardize image intensity values or scale to a uniform size and resolution.
- Techniques:** Pixel normalization (e.g., min-max scaling), resizing.
- Why:** Brings images into a consistent format, so OCR algorithms have predictable inputs.

OCR: Preprocessing

6.Binarization (Thresholding)

- Goal:** Convert images to only black and white pixels.
- Techniques:** Fixed threshold, Otsu's method or adaptive thresholding methods.
- Why:** Strong contrast between text and background optimizes the input for most OCR algorithms, which are designed to work on binary images.

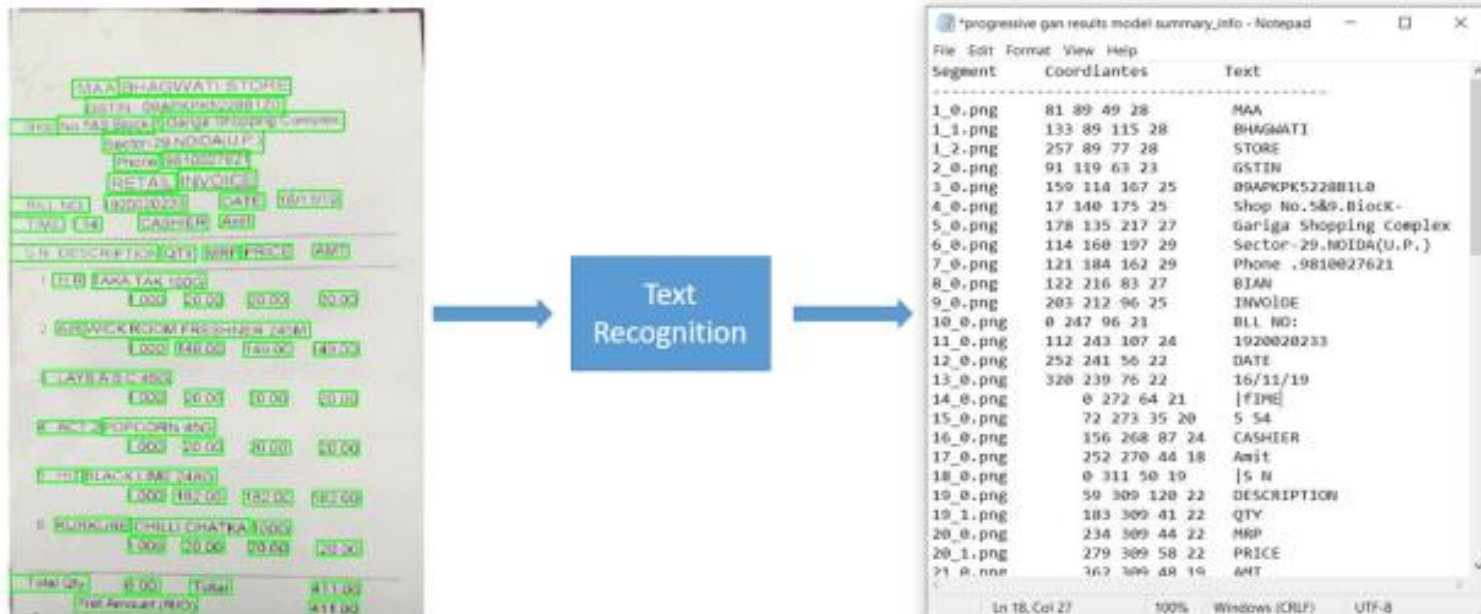
7.Inversion (If needed)

- Goal:** Convert black pixel into white and white into black. It means text and background colors are swapped.
 - Techniques:** Thresholding followed by inversion. Color to grey then thresholding convert it into binary then inversion.
- Why:** OCR engines are generally optimized to recognize text that appears as dark characters on a light background. When the input image has light-colored text on a dark background (such as white text on a black sign), inverting the image puts it in the preferred format (dark text on light background).

OCR: Preprocessing

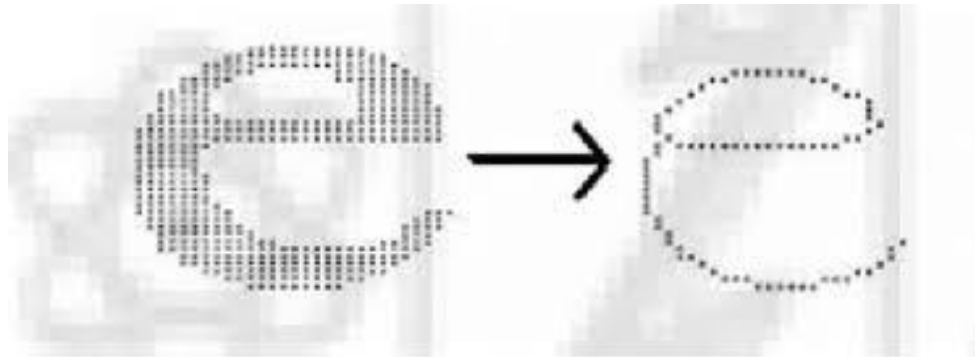
8. Text Detection, Layout, and Segmentation

- Goal:** Detect lines, words, or blocks of text and segment them from one another.
- Techniques:** Connected component analysis(utilize 4 connectivity or 8 connectivity to identify connected components), projection profiles, or deep learning-based text detectors.
- Why:** Helps the OCR system understand document structure and process multi-line/column/page layouts. This is a crucial step in complex document processing.



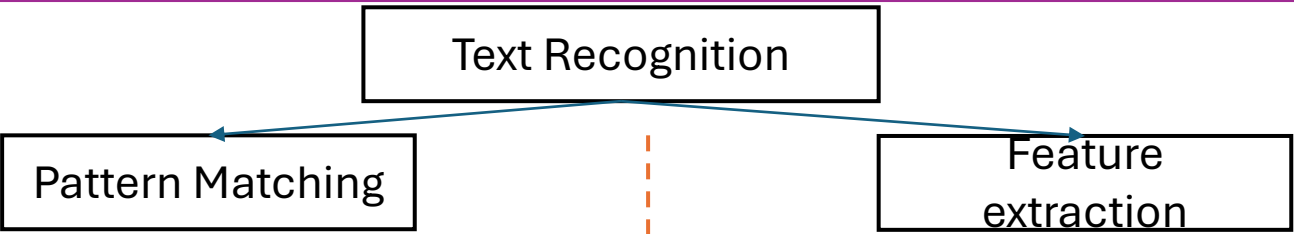
9. Thinning (Skeletonization)

- Goal:** Reduce character width to a single-pixel skeleton without losing shape.
 - Why:** Enhances the outline of handwritten or joined text, aiding recognition, especially in noisy or stylized scripts
- Techniques:** Morphological Based techniques like erosion etc.

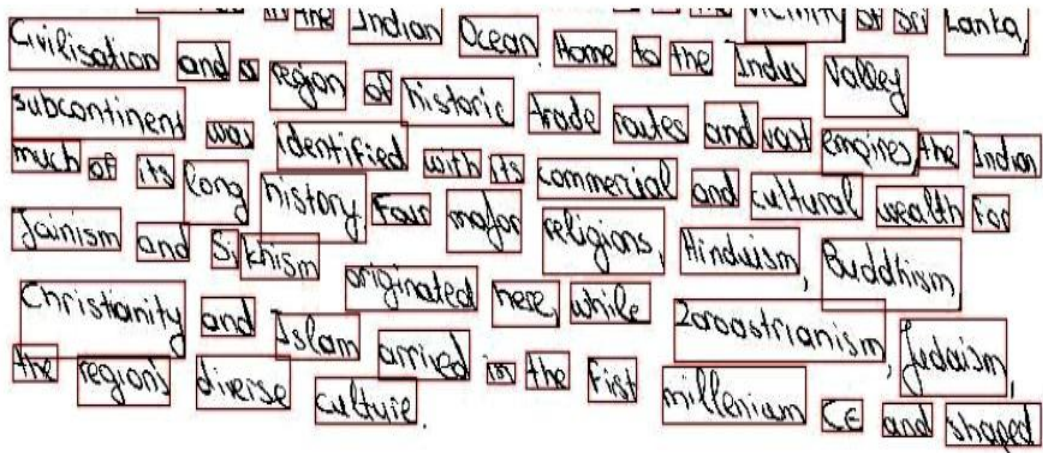
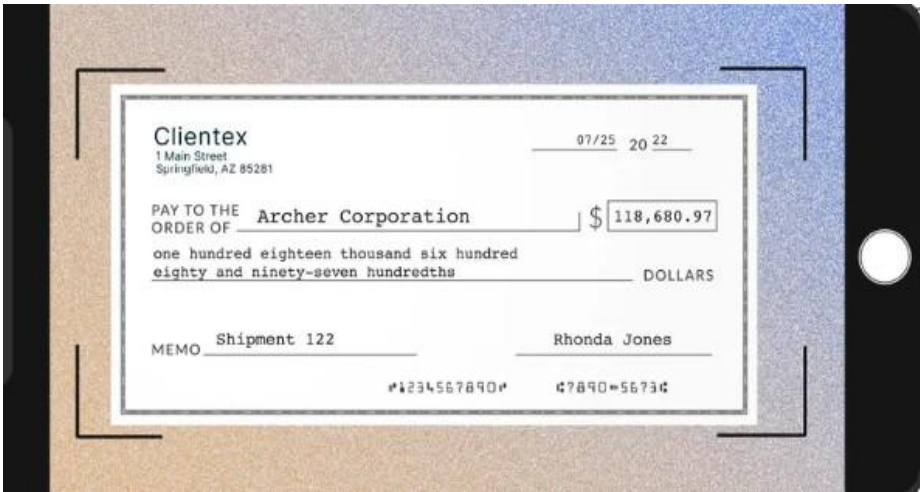


OCR: Text Recognition Techniques

Compares extracted character images (with stored templates.



Analyze geometrical features like edge and line etc.



Technique	Methodology	Best Use Cases	Limitations
Pattern Matching	Direct glyph comparison with templates.	Typed, uniform, printed fonts.	Poor with font/style/size variation, handwriting.
Feature Extraction	Analyzes structural features, uses ML to classify.	Handwriting, varied or complex fonts.	Requires feature engineering and more computation.

Note: glyph is the specific visual shape of a character within a particular font.

OCR: Flow diagram of Template matching

Image acquisition, Pre-processing, Segmentation

- Obtain scanned or captured image of the document.
- Convert image to grayscale or binary.
- Detect the ROI.

Normalization

- Resize each character segment to match template size.
- Perform normalization such as scaling, alignment.

Template Database Preparation

- Maintain a database of standardized character templates (A-Z, 0-9, etc.).

Template Matching

- For each segmented character, compute similarity (e.g., correlation or distance metric) with each template.
- Use matching functions such as Euclidean distance, Correlation etc.

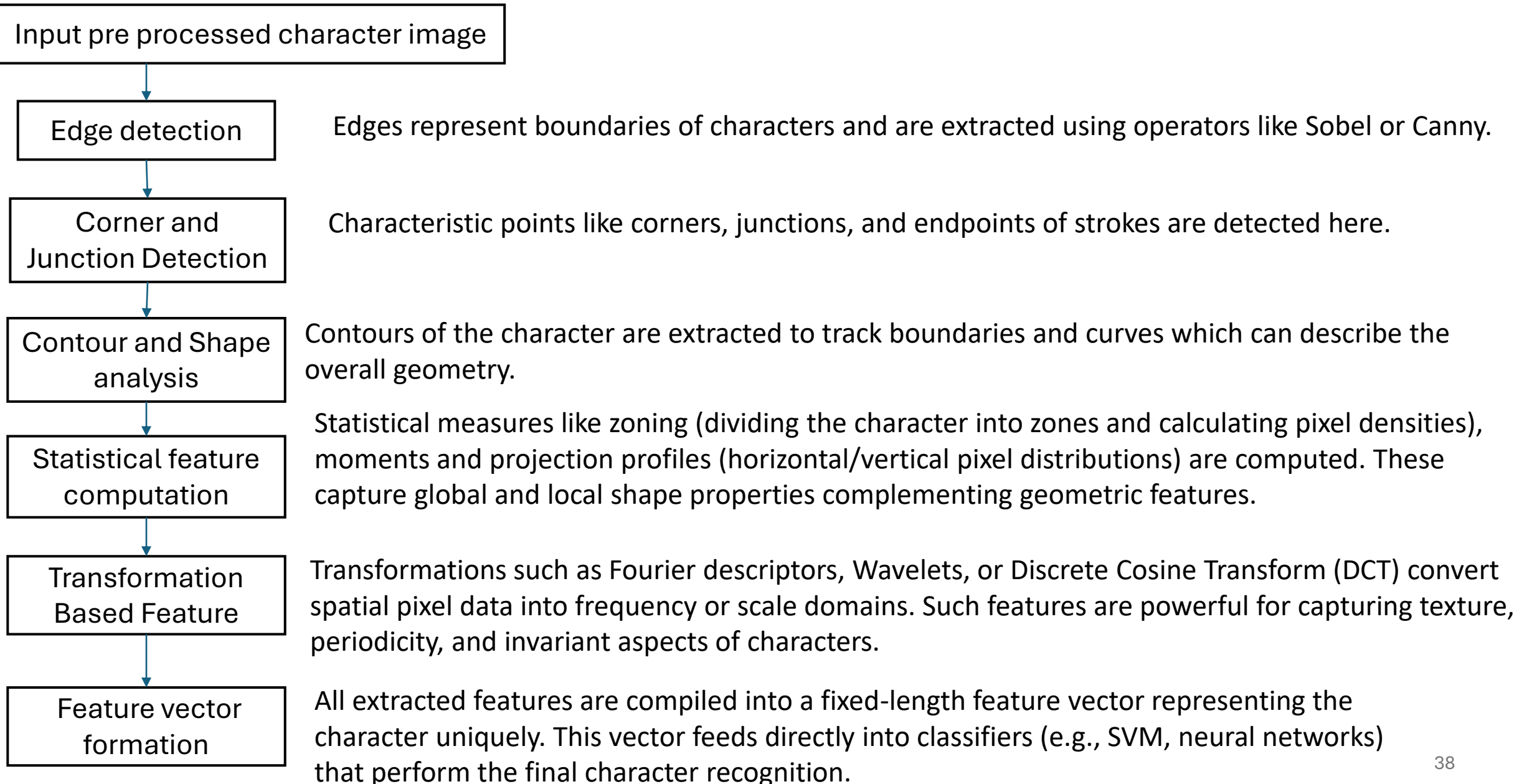
Best match selection

- Identify template with highest similarity or least error to input segment.
- Assign label of the best matched template as recognized character.

Post- Processing

- Aggregate recognized characters to form words/lines.
- Optionally, apply correction or validation with dictionary or context-aware modules.

OCR: Flow diagram of Feature extraction



OCR: Post Processing

The **first step** of post-processing is to **identify which words are likely wrong**.

- Method:**

- Typically uses a dictionary lookup (spell-checking) or analyzes the **confidence scores** provided by the OCR engine for each character/word.
- Words not found in the dictionary or those with low confidence are flagged as potential errors.

Second step: From the list of candidates, the **best correction is chosen based on the context of the sentence** or document.

- Method:** A language model or n-gram analysis is used to determine which candidate word is most probable given the words surrounding it.

- For example, if the OCR output is "The *house* was **on fre**",
- The language model will select "fire" over other candidates like "free" or "frog" to complete the sentence "The *house* was **on fire**."

Third Step: This step applies domain-specific rules to check and **correct structured information** that a dictionary can't validate.

- Method:** Uses regular expressions (RegEx) or other rules to ensure data fields

like dates,

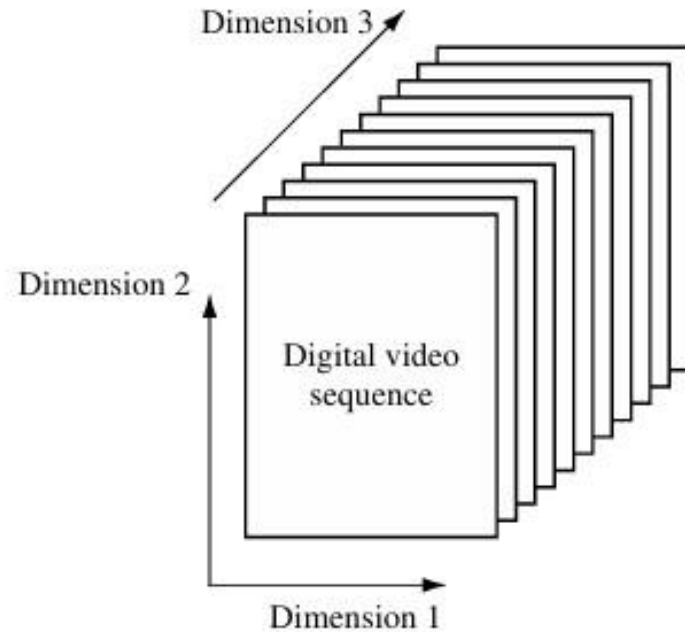
phone numbers,

currency amounts, and addresses adhere to the correct format (e.g., correcting a date "10-99-2025" to "10-09-2025").

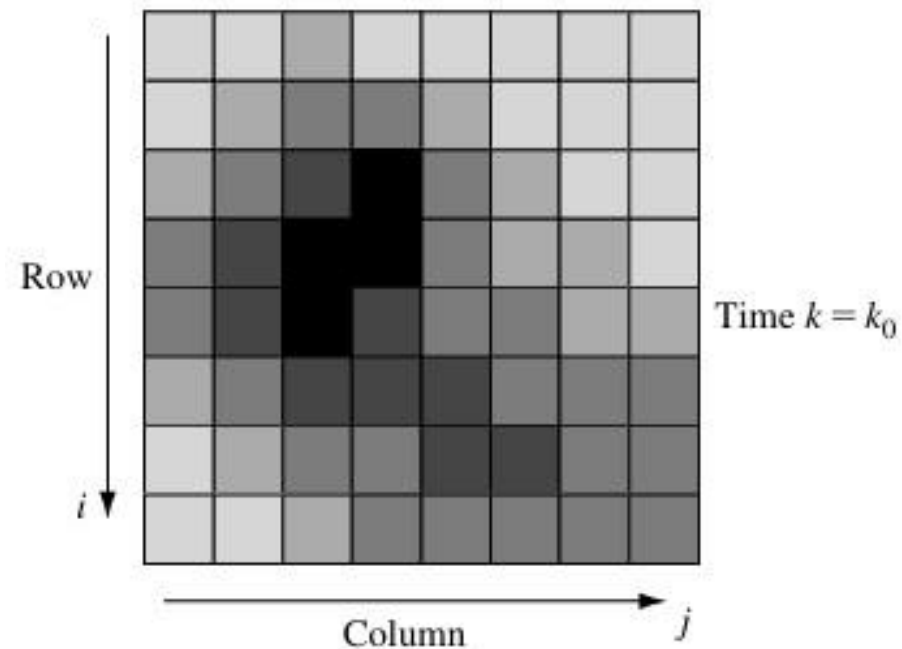
Video processing: Difference between image and video

Image: A single static frame, a snapshot of a moment.

Video: A sequence of images (frames) displayed at a certain frames per second (FPS) to show the motion.



Digital video is ordinarily a function of three dimensions—two in space and one in time,



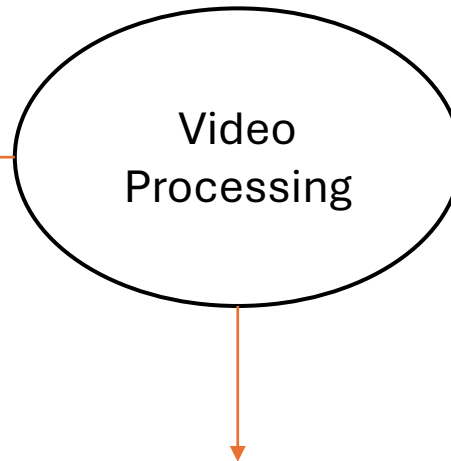
A single frame

Video processing: Difference between image and video

Image: A single static frame, a snapshot of a moment.

Video: A sequence of images (frames) displayed at a certain frames per second (FPS) to show the motion.

Video processing involves applying mathematical and computational techniques to video sequences (time-varying images) to extract information and improve quality.



Video can be understood as a time series of frames, and video processing manages both spatial information (pixel values within a frame) and temporal information (changes between consecutive frames).

Applications like object detection, motion tracking or streaming etc.

Video processing: Overall Flow diagram

Video acquisition
and formation



A video is formed by sequencing a series of still images, called frames, and displaying them rapidly to create the illusion of continuous motion.

This process is known as video formation and is grounded in the principle of persistence of vision, where the human eye perceives a fluid motion if frames are shown quickly enough—typically 24, 30, or 60 frames per second (fps).

Preprocessing



Video sampling,
Frame extraction,
Resizing,
Denoising and enhancement.

Motion estimation and object
tracking



analyze changes across frames to identify movement, track objects, or detect events.

Video
Compression



One of the most critical steps—used to remove spatial and temporal redundancies.

Standards like MPEG, H.264, and H.265 achieve efficient compression while preserving perceptual quality.

Video formation

A video is formed by sequencing a series of still images, called frames, and displaying them rapidly to create the illusion of continuous motion. This process is known as video formation and is grounded in the principle of persistence of vision, where the human eye perceives a fluid motion if frames are shown quickly enough—typically 24, 30, or 60 frames per second (fps).

- Capture:** A video camera or imaging sensor captures a scene at discrete time intervals. Each individual capture at a moment is one frame—essentially a static image.
- Storage:** These frames are then stored sequentially, initially as analog signals (in older systems) or immediately as digital data (in modern digital cameras).
- Rendering/Display:** When played back, these frames are displayed in rapid succession at a specified fps on a screen. The human brain blends these into a perception of smooth, continuous motion.
- Encoding/Compression:** Before a video is distributed or streamed, the sequence of frames is typically compressed using standards like MPEG, H.264, or HEVC. Compression leverages both spatial redundancy (within a frame) and temporal redundancy (between frames) to reduce file size.
- Playback:** On playback devices, these frames are decoded and rendered back into a video stream for viewers.

Video Sampling

Video sampling is the essential process of converting a continuous **analog video signal** into a **discrete digital signal**.

Spatial Sampling

Spatial sampling refers to selecting discrete points across the two spatial dimensions—width and height—of each video frame to create a grid of pixels.

Higher the spatial sampling rate, the more pixels per frame, as a result providing more detail.

A 1920x1080 HD frame is a result of sampling an image at 1920 horizontal and 1080 vertical points.

For high-resolution content, spatial sampling should satisfy the Nyquist criterion, ensuring the sampling frequency is at least twice the highest spatial frequency in the image to avoid loss of information and aliasing.

Temporal Sampling

Temporal sampling refers to how often the scene is sampled over time, essentially determining how many frames per second (fps) are captured. This converts continuous motion into a series of discrete snapshots.

Cameras or video systems sample the scene at uniform intervals—commonly 24, 30, or 60 times per second. Example: A standard film runs at 24 fps, meaning the scene is sampled 24 times every second.

Too low a temporal sampling rate causes motion artifacts like strobing or jerkiness (temporal aliasing).

Video Coding and Video Compression

Video Coding

- Video coding is the process of representing video data in a standardized digital format.
 - It converts uncompressed video (raw pixel data) into a bitstream following a specification like H.264, HEVC, or VP9.
 - It ensures video can be interpreted accurately by decoders across diverse devices and applications.
-

Video Compression

- Video compression is the reduction of video data size by removing redundant information and exploiting similarities within and between frames.
- Compression is achieved through codecs (COder/DECoder), which use coding methods to minimize file size while maintaining visual quality.
- Intraframe (Spatial) Compression: Compresses each frame individually by summarizing similar pixel regions—this addresses spatial redundancy.
- Interframe (Temporal) Compression: Stores only changes or differences between consecutive frames, exploiting temporal redundancy—this greatly improves compression efficiency for video streams.

Frame based Compression (MPEG)

Frame-based compression in MPEG is a method that reduces both spatial and temporal redundancy in a video by encoding and arranging frames of different types within a group of pictures (GOP).

Frame-based compression, as used in MPEG (Moving Picture Experts Group) standards, encodes video frame by frame.

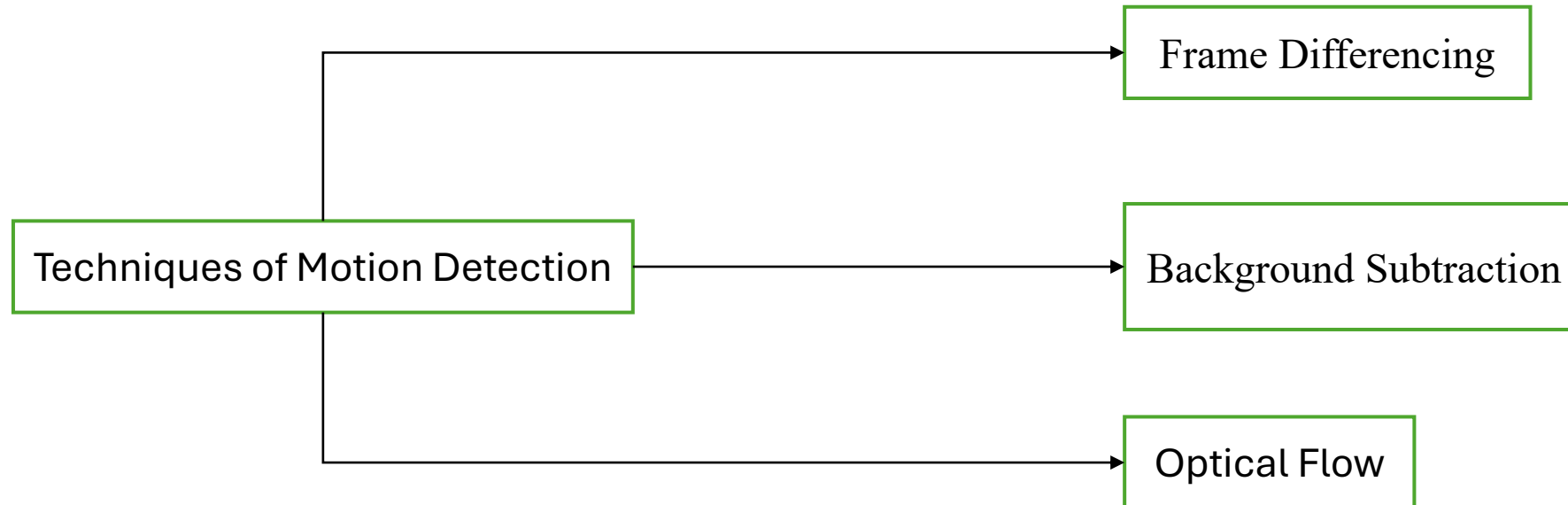
MPEG uses three types of frames:

- I-frames (Intra-coded): Encoded without reference to other frames.
- P-frames (Predicted): Encoded using data from previous frames.
- B-frames (Bi-directional): Encoded using both previous and future frames.

This combination allows MPEG to achieve high compression ratios while maintaining acceptable visual quality.

Motion Detection

1. Motion detection in image and video processing is a critical technique used to **identify changes in a sequence of images or video frames**. Essentially, it aims to determine if and where movement has occurred within a scene.
2. Motion detection focuses on analyzing temporal changes in pixel values across consecutive frames.



Motion Detection

Frame Differencing

1. This Method Involves subtracting one frame from another frame.
2. The Resulting difference highlight the areas where changes has been occurred.
3. It's computationally simple but sensitive to noise and lighting variations.

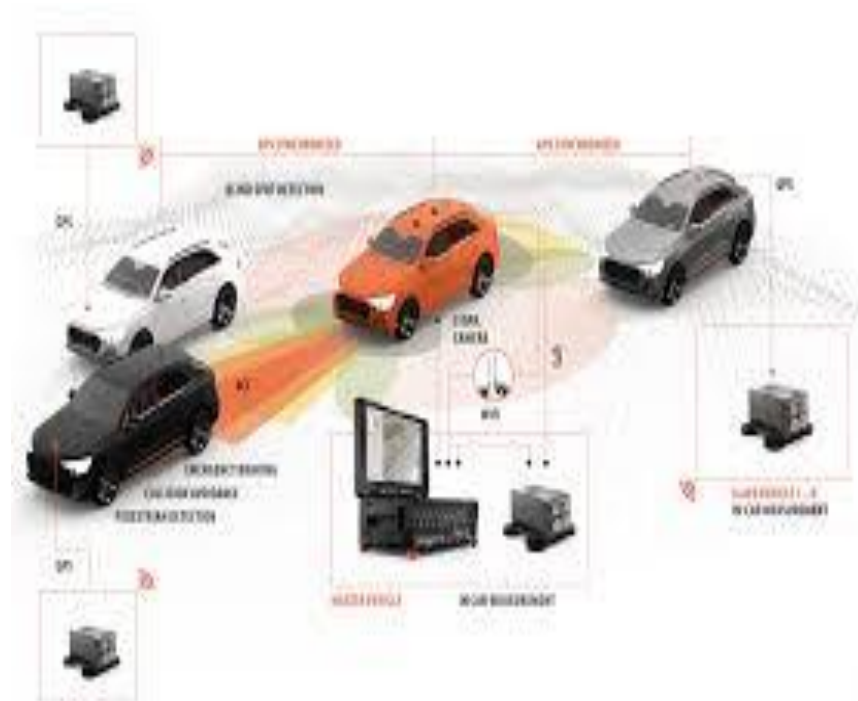
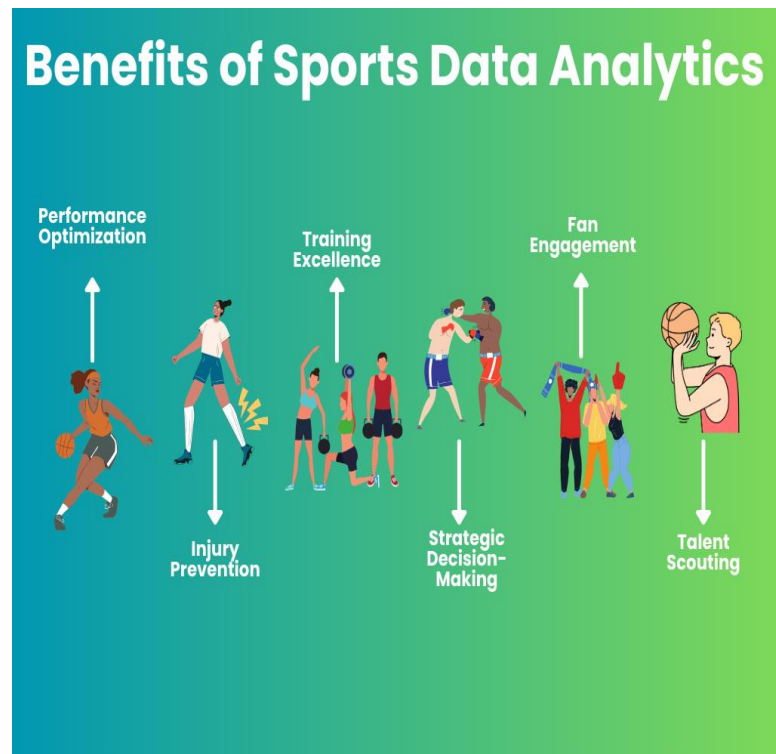
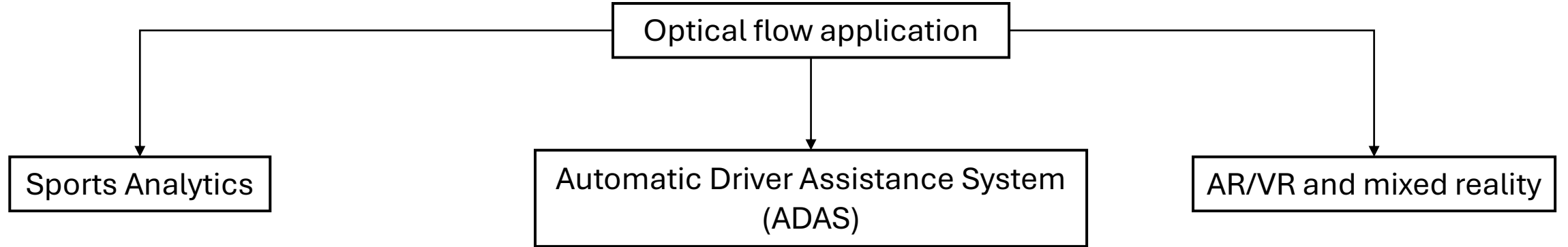
Background Subtraction

- 1.This technique involves creating a model of the static background and then subtracting it from each new frame.
- 2.The remaining pixels represent moving objects.
- 3.Background subtraction is more robust than frame differencing but requires a stable background model.

Optical Flow

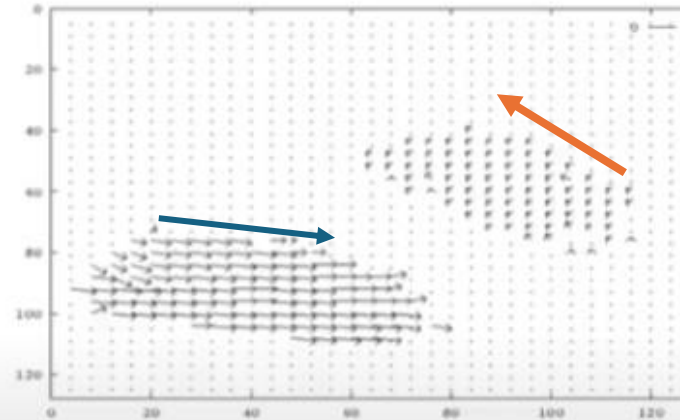
- 1.Optical flow estimates the apparent motion of objects between frames by analyzing the movement of pixels.
- 2.It provides a more detailed understanding of motion, including direction and velocity.
- 3.Optical flow is computationally intensive but can handle complex motion scenarios.

Optical Flow: Motivation or Applications



Optical flow: Definition and Explanation

- Optical flow refers to the pattern of apparent motion of brightness values across the image plane as a result of real-world motion, camera motion, or scene changes between consecutive frames.
- It basically measures, how image intensities move over time to reveal motion information such as object displacement, scene dynamics, and camera movement.
- **Motion of brightness patterns in an image is known as optical flow.**



For each pixel we compute the vector.

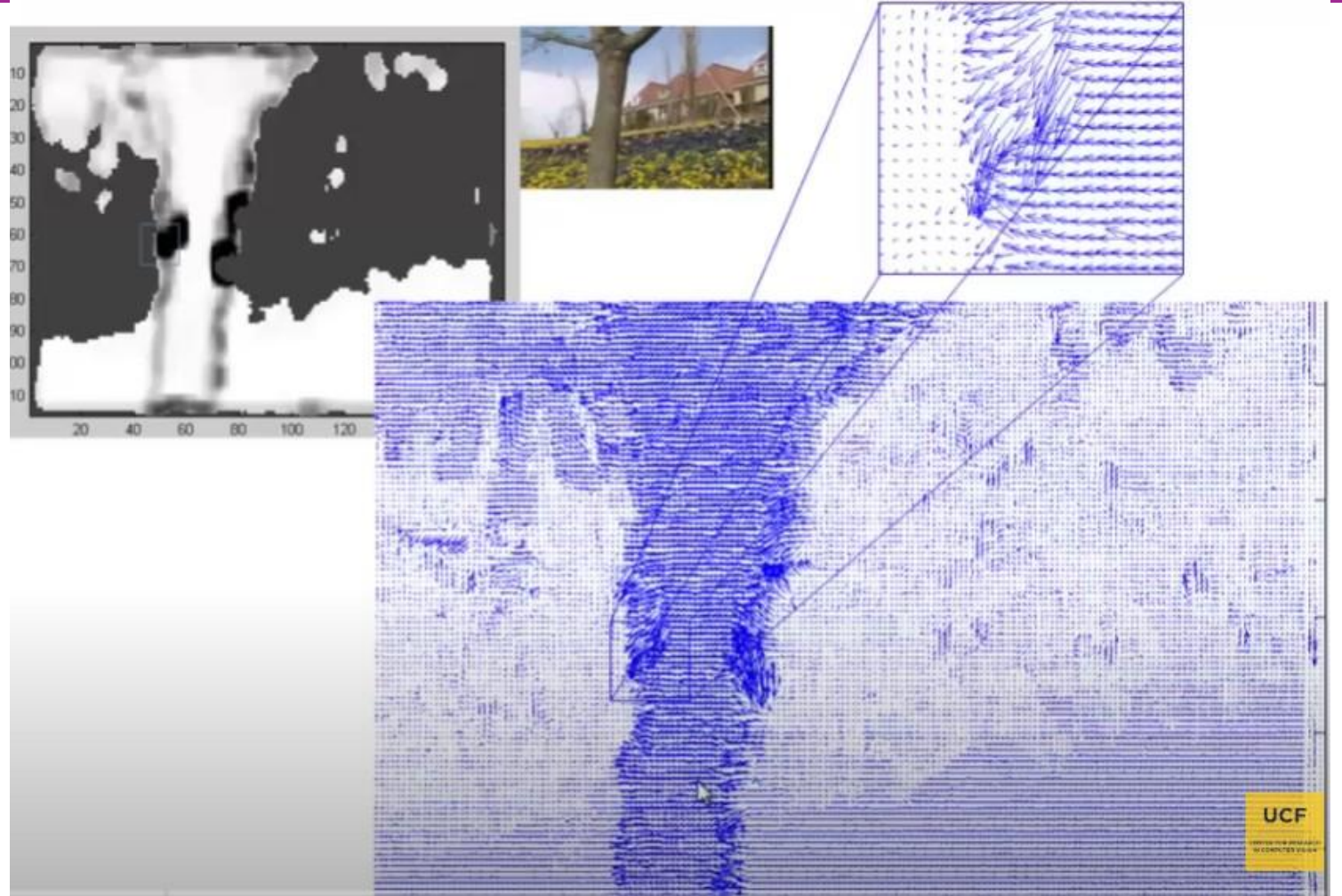
Based on it , we could identify how much that vector changed as compared to the previous frame.

Displacement vector is nothing but optical flow vector.

Optical flow: Definition and Explanation

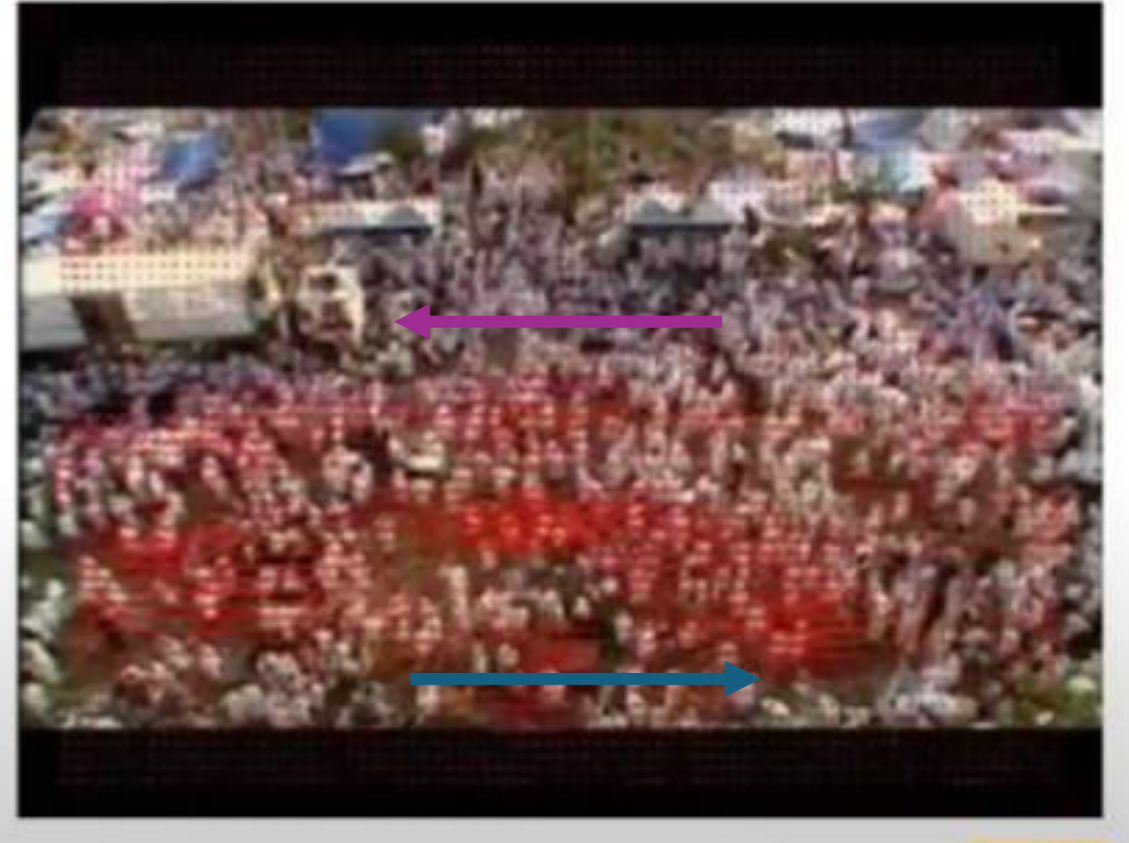


Every pixel is a vector.



At boundary there is always problem to identify the correct vectors, because there is discontinuity at the boundary.

Optical flow: Definition and Explanation



Crowded sequence where some group of pixels are moving in one direction and some other group of pixels are moving in other direction.

Optical flow: Definition and Explanation



Color coded optical flow instead of vectors. Here the different colors helps to identify the direction of motion.



Color coded Traffic Sequence.

Optical Flow: Measurement using constraint equation

Image taken at time t



t



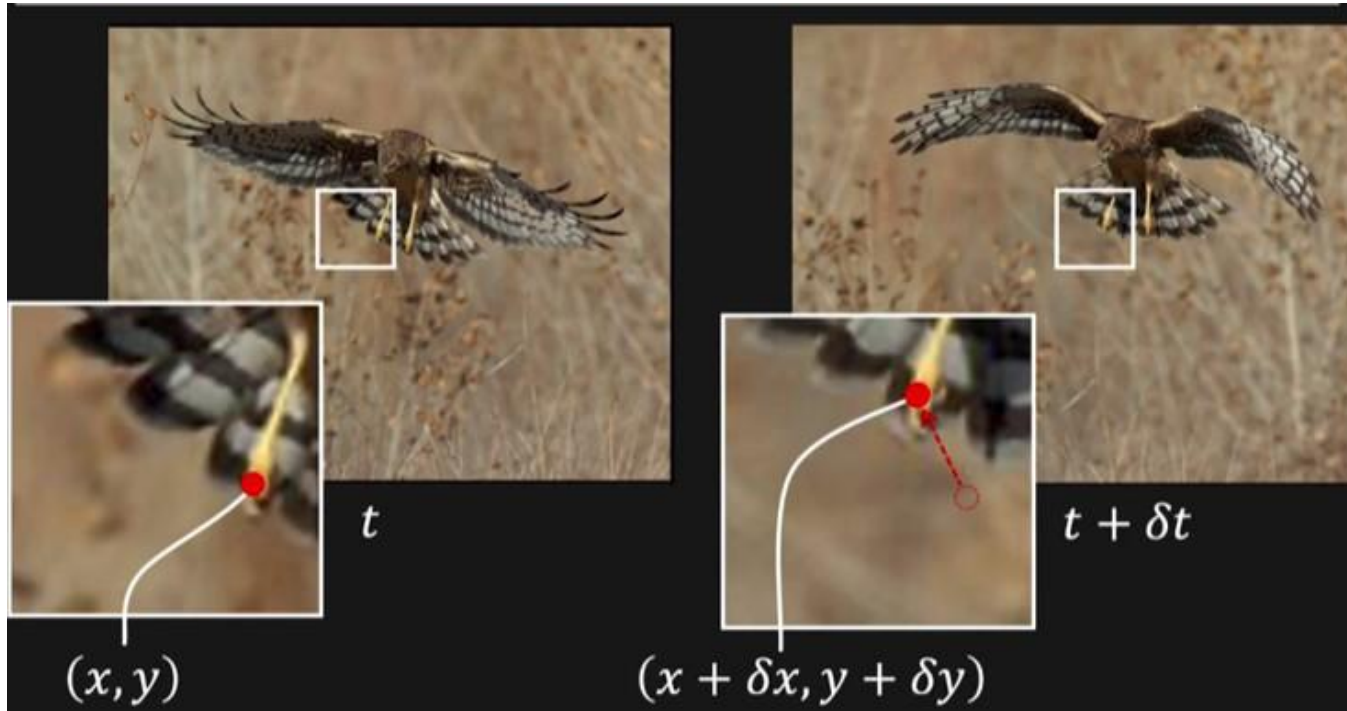
$t + \delta t$

Image taken at time $t + \delta t$.
Where δt is very small.



Consider a window in both the images
and concentrate on a single point in
the image.

Optical Flow: Measurement using constraint equation



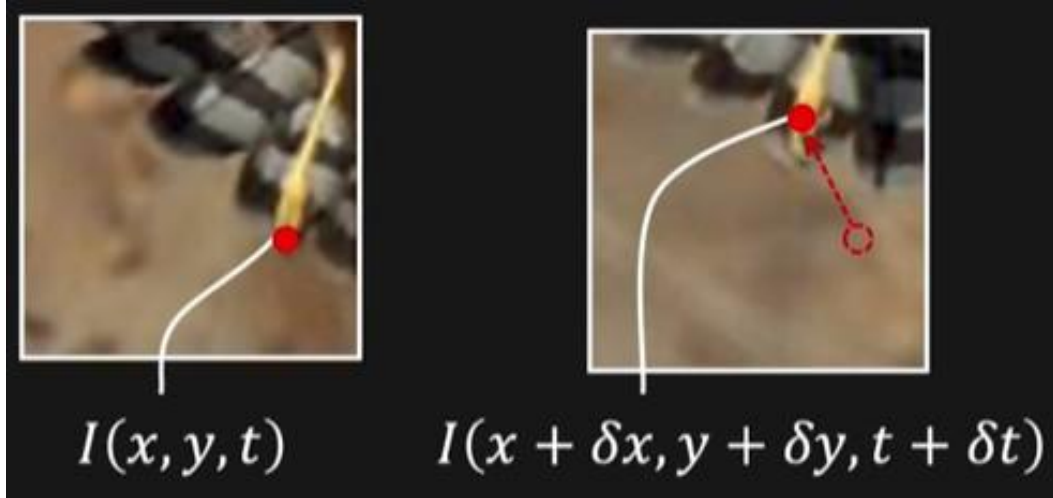
Consider a window in both the images and concentrate on a single point in Image.

At time $t + \delta t$, a point (x, y) is moved to the new location i.e. $(x + \delta x, y + \delta y)$.

Displacement: $(\delta x, \delta y)$ Optical Flow: $(u, v) = \left(\frac{\delta x}{\delta t}, \frac{\delta y}{\delta t} \right)$

Here, (u, v) is the speed of the point in x and y direction i.e. known as optical flow and this is what we are going to measure.

Optical Flow: Measurement using constraint equation



For measuring the same, we need to make some assumptions.

The first assumption is, Brightness of image points remains constant over time.

(At least for two consecutive frames.)

$$I(x + \delta x, y + \delta y, t + \delta t) = I(x, y, t)$$

Second assumption is displacement and time steps are very small.

Displacement $(\delta x, \delta y)$ and time step δt are small

Next is, approximation of the intensities. By using Taylor series expansion.

Expand a function as an infinite sum of its derivatives

$$f(x + \delta x) = f(x) + \frac{\partial f}{\partial x} \delta x + \frac{\partial^2 f}{\partial x^2} \frac{\delta x^2}{2!} + \dots + \frac{\partial^n f}{\partial x^n} \frac{\delta x^n}{n!}$$

If δx is small:

$$f(x + \delta x) = f(x) + \frac{\partial f}{\partial x} \delta x + \boxed{O(\delta x^2)} \rightarrow \text{Almost Zero}$$

For a function of three variables with small $\delta x, \delta y, \delta t$:

$$f(x + \delta x, y + \delta y, t + \delta t) \approx f(x, y, t) + \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial t} \delta t$$

Rewrite the above equation in this way -

$$I(x + \delta x, y + \delta y, t + \delta t) = I(x, y, t) + I_x \delta x + I_y \delta y + I_t \delta t$$

Optical Flow: Measurement using constraint equation

Our two assumptions, brightness assumption and displacement assumptions provide us the following two equations

$$I(x + \delta x, y + \delta y, t + \delta t) = I(x, y, t) \quad \text{----- (1)}$$

Brightness Assumption

$$I(x + \delta x, y + \delta y, t + \delta t) = I(x, y, t) + I_x \delta x + I_y \delta y + I_t \delta t \quad \text{----- (2)}$$

Displacement Assumption

$$\text{Subtract (1) from (2): } I_x \delta x + I_y \delta y + I_t \delta t = 0$$

$$\text{Divide by } \delta t \text{ and take limit as } \delta t \rightarrow 0: I_x \frac{\partial x}{\partial t} + I_y \frac{\partial y}{\partial t} + I_t = 0$$

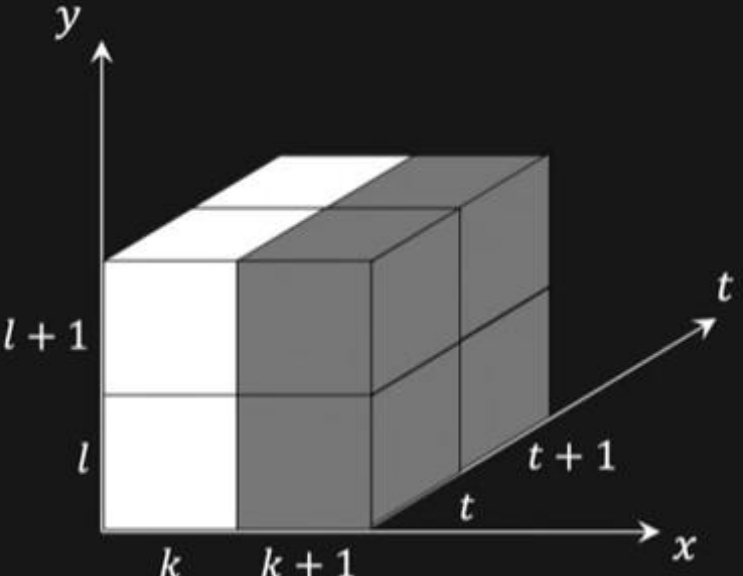
Constraint Equation: $I_x u + I_y v + I_t = 0$ (u, v) : Optical Flow

(I_x, I_y, I_t) can be easily computed from two frames

How to compute I_x , I_y and I_t ?

Optical Flow: Measurement using constraint equation

For I_x , Subtract one half portion (black) from the other half portion (white).


$$I_x(k, l, t) = \frac{1}{4}[I(k+1, l, t) + I(k+1, l, t+1) + I(k+1, l+1, t) + I(k+1, l+1, t+1)] - \frac{1}{4}[I(k, l, t) + I(k, l, t+1) + I(k, l+1, t) + I(k, l+1, t+1)]$$

I_x used to determine, how much change in the intensity in x direction. Same can be done for I_y and I_t .

Optical Flow: Measurement using constraint equation

Geometrical Interpretation of optical flow constraint equation.

For any point (x, y) in the image, its optical flow (u, v) lies on the line:

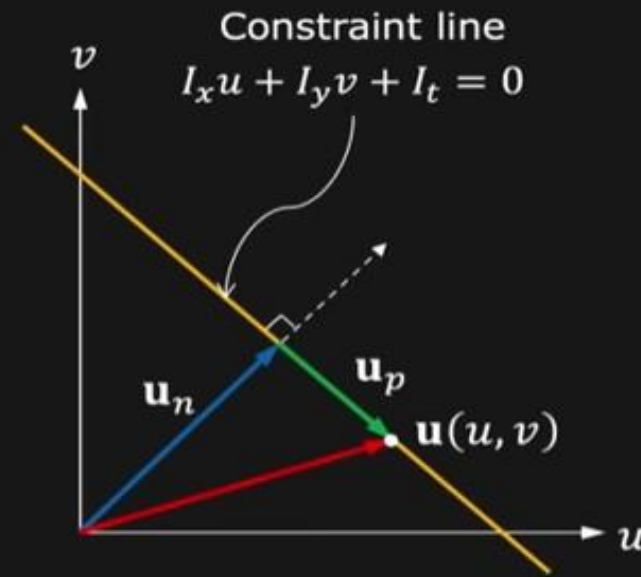
$$I_x u + I_y v + I_t = 0$$

Optical Flow can be split into two components.

$$\mathbf{u} = \mathbf{u}_n + \mathbf{u}_p$$

$\mathbf{u}_n$: Normal Flow

$\mathbf{u}_p$: Parallel Flow



For any point (x, y) , its optical flow (u, v) lies on the line but where it is lying that makes this problem as a under constrained problem and it split it into two parts one is normal component and other one is parallel components.

Here, under constrained problem means, more unknown as compared to the number of equations.

As a results, many infinite solution, because available information is insufficient to determine the unique solution.

Constraint Equation: $I_x u + I_y v + I_t = 0$

2 unknowns, 1 equation.

What can be computed from here -

It is to find that, Normal flow can be computed from here but parallel flow is difficult to find because it may be anywhere to line.

Optical Flow: Measurement using constraint equation

Normal flow can be computed just from the constraint line.

Direction of Normal Flow:

Unit vector perpendicular to the constraint line:

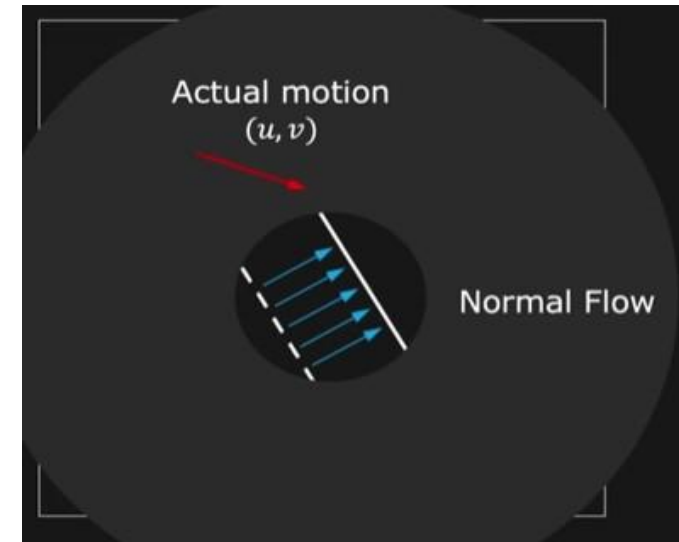
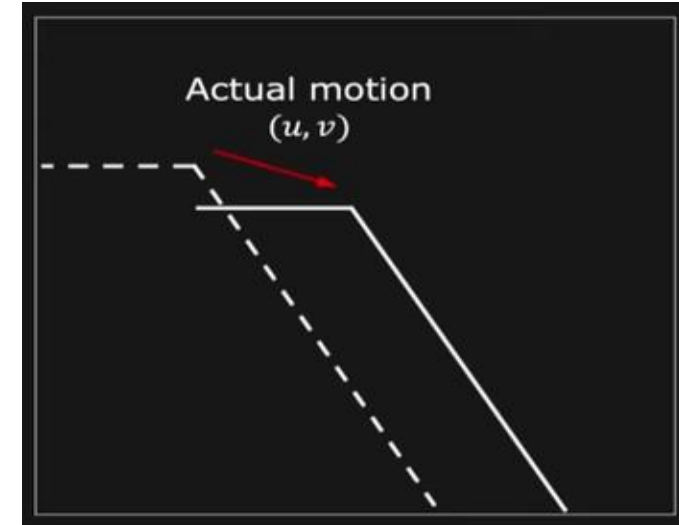
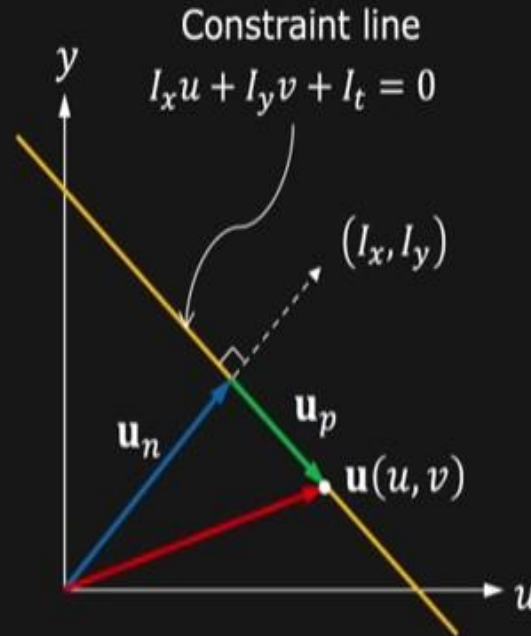
$$\hat{\mathbf{u}}_n = \frac{(I_x, I_y)}{\sqrt{I_x^2 + I_y^2}}$$

Magnitude of Normal Flow:

Distance of origin from the constraint line:

$$|\mathbf{u}_n| = \frac{|I_t|}{\sqrt{I_x^2 + I_y^2}}$$

$$\mathbf{u}_n = \frac{|I_t|}{(I_x^2 + I_y^2)} (I_x, I_y)$$



Looking from an aperture, we can say its normal flow.

Optical Flow: Lucas Kanade method

Overall what we have, 2 unknowns and 1 equations. For solving the same, some additional equations or constraint or assumptions are required.

Assumption: For each pixel, assume Motion Field, and hence Optical Flow (u, v) , is constant within a small neighborhood W .



That is for all points $(k, l) \in W$:

$$I_x(k, l)u + I_y(k, l)v + I_t(k, l) = 0$$

Assumption is, all the pixels are moving in the same direction, or we can say their motion field are same and hence optical flow is same.

Consider a small window (w) and from that small window, consider a point (k, l) . If motion field is constant, it means derivative of intensity in the x direction, derivatives of intensities in the y direction and same in time direction are equal to zero.

Optical Flow: Lucas Kanade method

For all points $(k, l) \in W$: $I_x(k, l)u + I_y(k, l)v + I_t(k, l) = 0$

Let the size of window W be $n \times n$

In matrix form:

$$\begin{bmatrix} I_x(1,1) & I_y(1,1) \\ I_x(k,l) & I_y(k,l) \\ \vdots & \vdots \\ I_x(n,n) & I_y(n,n) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} I_t(1,1) \\ I_t(k,l) \\ \vdots \\ I_t(n,n) \end{bmatrix}$$

A $\mathbf{u}$ B
 (Known) (Unknown) (Known)
 $n^2 \times 2$ 2×1 $n^2 \times 1$

n^2 Equations, 2 Unknowns: Find Least Squares Solution.

For lot of pixels inside the window (W), there will be a matrix.

Number of equations is directly equal to the number of pixels inside the window.

This solution works because all the linear equations are independent.

Solve linear system: $A\mathbf{u} = B$

$$A^T A \mathbf{u} = A^T B \quad (\text{Least-Squares using Pseudo-Inverse})$$

In matrix form:

$$\begin{bmatrix} \sum_W I_x I_x & \sum_W I_x I_y \\ \sum_W I_x I_y & \sum_W I_y I_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} -\sum_W I_x I_t \\ -\sum_W I_y I_t \end{bmatrix}$$

$A^T A$ $\mathbf{u}$ $A^T B$
 (Known) (Unknown) (Known)
 2×2 2×1 2×1

$$\mathbf{u} = (A^T A)^{-1} A^T B$$

Fast and Easy to Solve

All the values are from the spatial domain.

All the values are from the time domain.

Indices (k, l) not written for simplicity

When Does Optical Flow Estimation Work?

$$A\mathbf{u} = B$$

$$A^T A\mathbf{u} = A^T B$$

- $A^T A$ must be **invertible**. That is $\det(A^T A) \neq 0$
- $A^T A$ must be **well-conditioned**.

If λ_1 and λ_2 are eigen values of $A^T A$, then

$$\lambda_1 > \epsilon \text{ and } \lambda_2 > \epsilon$$

$$\lambda_1 \geq \lambda_2 \text{ but not } \lambda_1 \gg \lambda_2$$

Here,
well conditioned means, a significant change in the output with a small change in the input.

If system is well conditioned, then only
You will find the robust solution for the optical flow.

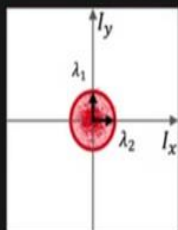
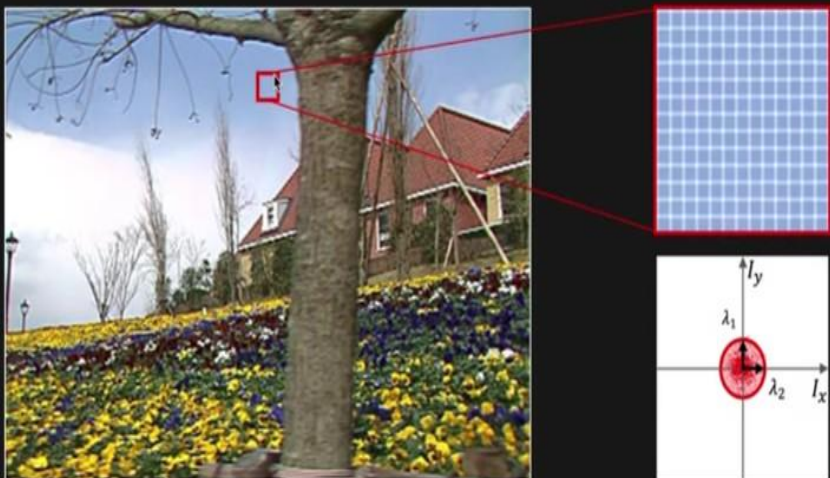
As we have 2 x 2 matrix, as a result
2 eigen values.

Eigen value 1 and Eigen value 2 both Should
be significantly large.

At the same time, If one eigen value is high
and it is significantly larger than other one,
that may also create the problem for optical
flow detection.

Optical Flow: Lucas Kanade method

Smooth Regions (Bad)



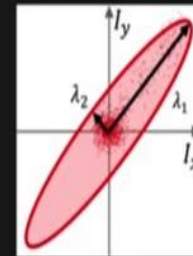
$\lambda_1 \sim \lambda_2$
Both are Small

Equations for all pixels in window are more or less the same

Cannot reliably compute flow!

- All the gradients are very small. The eigen values are very small.
- No texture.
- If we move the window, then optical flow can not estimate reliably.

Edges (Bad)

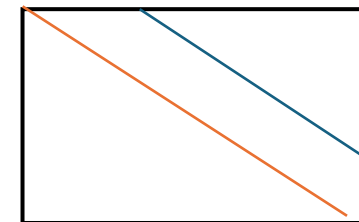


$\lambda_1 \gg \lambda_2$

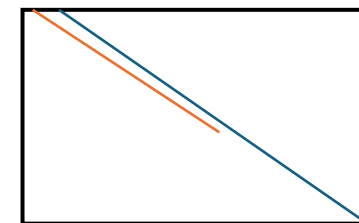
Badly conditioned. Prominent gradient in one direction.

Cannot reliably compute flow!
Same as Aperture Problem.

- Eigen value in one direction is small, and it is large in the other direction.



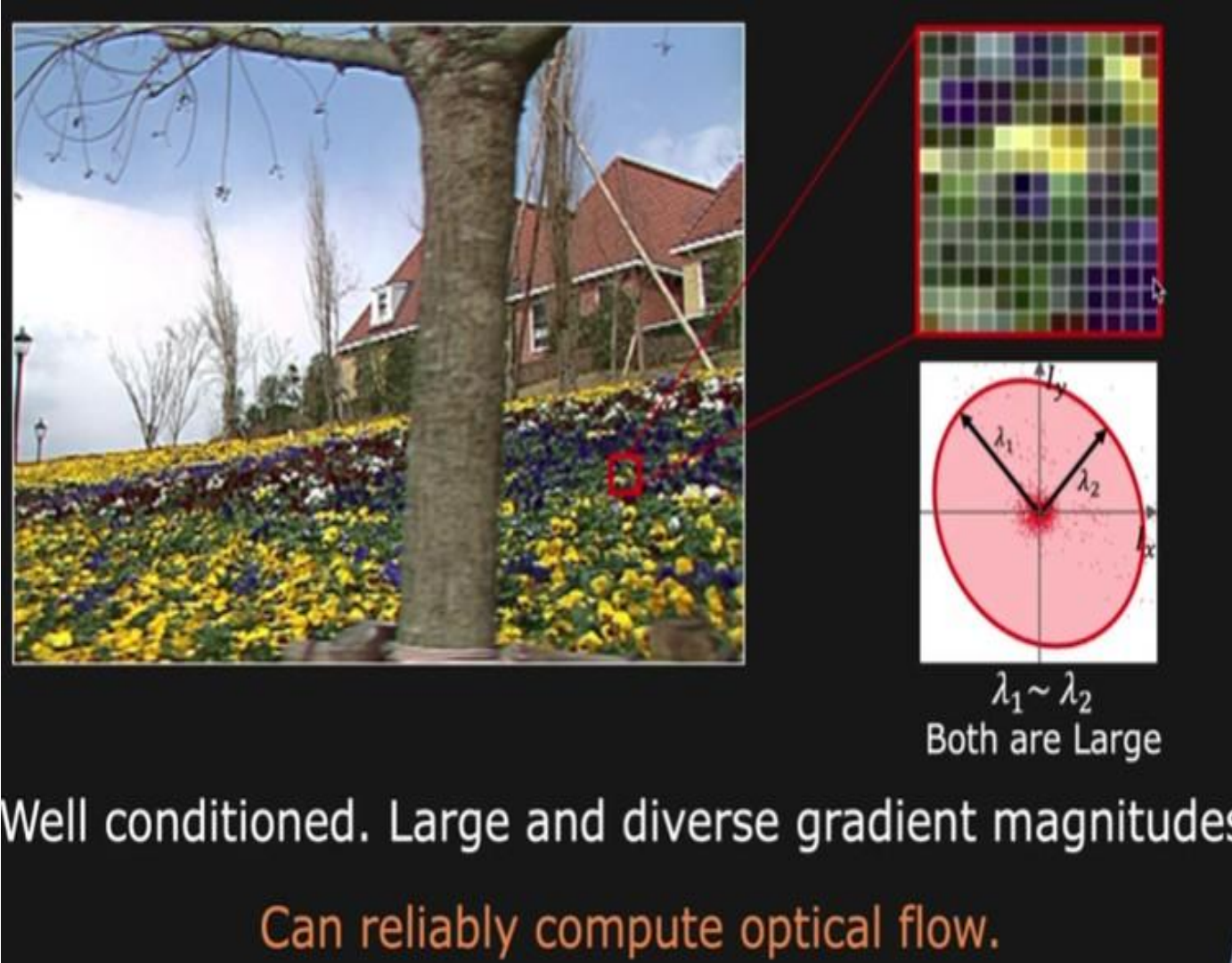
here, edge can move up and down



it may increase diagonally.

In both the above cases, the difference between pixels are not so significant.

Optical Flow: Lucas Kanade method



Third case, where the eigen values are large.
It will help us to get diverse magnitude and at the
Same time optical flow can be reliably computed.

Optical Flow: Problem and solution

What if we have Large Motion?



Taylor Series approximation of
 $I(x + \delta x, y + \delta y, t + \delta t)$ is not valid



Our simple linear
constraint equation not valid

$$I_x u + I_y v + I_t \neq 0$$

1. Lucas Kanade work well for small motion.
2. If the object moves faster, the brightness changes rapidly.

In this scenario 2 x 2 and 3 x 3 mask fail to estimate spatiotemporal derivatives.

3. Pyramids can be used to compute large optical flow vectors. (It takes large motion and reduces the motion)

Optical Flow: Problem and Solution.

Resolution pyramid, like fine to coarse resolution can be used to find out the large optical flow.

Steps followed -

Select 2 x 2 window,

For each 2 x 2 window find the average and use that value in new low-resolution image.



At some points, all motions are going to be less than a pixel. In that scenario, optical flow equation becomes valid.